

Causal Inference in Reinforcement Learning

B. Tech. Project

Submitted in partial fulfillment of the requirements

of the degree of

Bachelor of Technology

in

Electrical Engineering

By

Jasmer Singh Sanjotra

(220002041)

Under the guidance of

Prof. Dibbendu Roy



Department of Electrical Engineering

Indian Institute of Technology Indore

Dec. 2025

Declaration

I hereby declare that the project entitled “Causal Inference in Reinforcement Learning” submitted in partial fulfillment for the award of the degree of Bachelor of Technology in Electrical Engineering completed under the supervision of Dr. Dibbendu Roy, Assistant Professor, Department of Electrical Engineering, Indian Institute of Technology Indore is an authentic work.

Further, I declare that I have not submitted this work for the award of any other degree elsewhere.

I declare that this written submission represents my ideas in my own words and where other’s ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misinterpreted or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Jasmer Singh Sanjotra

Date:

Place:

Preface

This report, titled “Causal Inference in Reinforcement Learning” has been prepared under the guidance of Prof. Dibbendu Roy as part of my Bachelor’s Thesis Project.

The objective of this work is to investigate how explicit causal reasoning, through the framework of Structural Causal Models can enhance the statistical efficiency, stability, and theoretical guarantees of policy-gradient reinforcement learning algorithms. The report presents a comprehensive treatment of the underlying theory, the motivation for counterfactual baselines, the methodological contributions developed during the project, and an extensive empirical evaluation across both synthetic and applied environments.

Every effort has been made to articulate the concepts clearly, maintain mathematical rigour, and present the material in a structured and accessible manner. Figures, diagrams, and experimental visualizations have been included wherever helpful to enhance interpretability. I hope that this work serves as a useful reference for students and researchers interested in the intersection of causality and reinforcement learning.

I sincerely thank Prof. Roy for his guidance, support, and insightful discussions throughout the course of this project.

Jasmer Singh Sanjotra

B.Tech. IV Year

Department of Electrical Engineering

B. Tech. Project Approval Certificate

This is to certify that the dissertation titled “**Causal Inference in Reinforcement Learning**” submitted by **Jasmer Singh Sanjotra** (Roll No. 220002041) is approved for the award of degree of **Bachelor of Technology** in **Electrical Engineering**.

BTP Supervisor(s):

Date:

Designation:

Acknowledgements

I would like to express my sincere gratitude to my project supervisor, Prof. Dibbendu Roy, for his unwavering support, insightful guidance, and constant encouragement throughout the duration of this work. His expertise and mentorship have played an invaluable role in shaping the direction and quality of this project. It has been a privilege to work under his supervision.

I extend my heartfelt thanks to my family for their unconditional love, patience, and belief in me. Their continuous support has been my strongest source of strength throughout my academic journey.

I am deeply grateful to my friends, whose encouragement, camaraderie, and timely motivation kept me grounded and focused during challenging phases of the project.

I would also like to acknowledge the broader Reinforcement Learning and Causality research community, including researchers and practitioners whose discussions, open-source work, and online contributions have significantly helped deepen my understanding of the field. In particular, the foundational work of thinkers such as Judea Pearl has greatly inspired the causal perspective developed in this thesis.

To all who have supported me, directly or indirectly, I express my sincere thanks.

Abstract

Policy-gradient algorithms are a foundational component of reinforcement learning (RL), yet they suffer from high variance in gradient estimates, especially in stochastic environments. This variance hampers sample efficiency and slows convergence. Many RL problems, however, possess latent causal structure in which different actions share common exogenous noise. This presents an opportunity that classical baselines do not exploit.

This thesis leverages Structural Causal Models (SCMs) to construct *counterfactual baselines* that evaluate alternative actions under the same exogenous disturbances. These baselines isolate the true action effect and significantly reduce estimator variance. We develop five such baselines, analyze their variance properties, and establish regret bounds showing that variance reduction directly improves sample complexity. We also introduce *C-PPO*, a causal extension of Proximal Policy Optimization that combines PPO’s stability with SCM-derived counterfactual advantages.

Experiments across SCM-bandit settings and a queueing MDP demonstrate substantial improvements: up to 98% reduction in gradient variance, consistent regret improvements, and as much as 50% reduction in sample complexity for C-PPO. These results highlight the practical and theoretical value of integrating causal reasoning into modern policy-gradient methods.

Keywords: Reinforcement Learning, Policy Gradients, Structural Causal Models, Counterfactual Baselines, Variance Reduction, Proximal Policy Optimization.

Contents

1	Introduction	1
1.1	Mathematical Formulation of RL	1
1.2	Regret and Sample Complexity	2
1.3	Policy Gradient Methods	2
1.3.1	Baselines as Variance-Reduction Mechanisms	3
1.4	Motivation for Counterfactual Baselines	4
1.5	Overview	4
2	Literature Review	5
2.1	Related Works	5
2.1.1	Reinforcement Learning and Policy-Gradient Variance	5
2.1.2	Causality, SCMs, and Counterfactual Reasoning in RL	5
2.1.3	RL for Queueing and Communication Networks	6
2.1.4	Positioning of This Work	6
3	Background	7
3.1	Introduction	7
3.2	Structural Causal Models	7
3.2.1	Conceptual Description	7
3.2.2	Mathematical Formulation	8
3.2.3	Graphical Representation	9
3.3	Interventions and the Do-Operator	9
3.3.1	Conceptual Description	9
3.3.2	Formal Definition	9
3.4	Counterfactuals and the Abduction–Action–Prediction Procedure	10
3.4.1	Conceptual Description	10
3.4.2	Formal Framework	10

3.5	Do-Calculus	11
3.5.1	Conceptual Description	11
3.5.2	Formal Rules	11
3.6	SCMs in Reinforcement Learning Contexts	12
3.6.1	Conceptual Description	12
3.6.2	Formal Integration	12
3.7	Summary	12
4	Theoretical Propositions	13
4.1	Regret Bounds Under Variance Dependence	13
4.1.1	One-Step Expected Improvement	14
4.1.2	Bounding the Sum of Gradient Norms	15
4.1.3	Regret Bound	15
4.2	Variance Reduction via Counterfactual Baselines	17
4.2.1	Counterfactual Baseline Estimator	18
4.2.2	Variance Reduction via Decomposition of the Gradient Estimator	18
5	Proposed Methodology	22
5.1	Notation and Preliminaries	22
5.2	Baseline Definitions and Properties - C-REINFORCE	23
5.2.1	Baseline 1: Mean of all non-taken actions	23
5.2.2	Baseline 2: Policy-weighted mean of non-taken actions	23
5.2.3	Baseline 3: Top- k mean (unweighted)	24
5.2.4	Baseline 4: Top- k policy-weighted mean	24
5.2.5	Baseline 5: Top- k mean of $\pi \cdot R$ (probability-weighted reward ranking)	25
5.2.6	Variance-Reduction Intuition and Comparative Considerations	25
5.2.7	Implementation Details and Numerical Considerations	26
5.2.8	Integration with Policy-Gradient Updates	26
5.3	Causal Proximal Policy Optimization (C-PPO)	27
5.4	Summary	29
6	Experimental Setups	30
6.1	SCM-Bandit Environment	30
6.1.1	Overview	30
6.1.2	SCM Generation Procedure	31

6.1.3	Structural Equation Assignment	31
6.1.4	Noise Distributions	32
6.1.5	Action Space and Interventions	32
6.1.6	Reward Computation	33
6.1.7	Reproducibility	33
6.2	Parallel Queue SCM Environment	33
6.2.1	Overview	33
6.2.2	System Structure	34
6.2.3	Exogenous Noise Specification	34
6.2.4	Service Dynamics	34
6.2.5	Arrival and Routing Dynamics	35
6.2.6	Reward Definition	35
6.2.7	Interventions	35
6.2.8	Observation Model	35
6.2.9	Environmental Determinism Under Fixed Noise	36
6.3	Summary	36
7	Results and Analysis	37
7.1	SCM-Bandit Environment	37
7.1.1	Environment A: 6 Nodes, Connectivity 0.6	37
7.1.2	Environment B: 7 Nodes, Connectivity 0.3	39
7.1.3	Environment C: 7 Nodes, Connectivity 0.6	40
7.1.4	Cross-Environment Comparative Analysis	42
7.1.5	Summary	42
7.2	Results in the Queueing SCM Environment	43
7.2.1	Cumulative Regret Over Time	43
7.2.2	Sample Complexity Analysis	44
7.2.3	Discussion	46
8	Conclusion and Future Work	47

Chapter 1

Introduction

Introduction

Reinforcement Learning (RL) is a formal framework for sequential decision-making under uncertainty. An RL problem is characterized by an agent interacting with an environment over discrete time steps, selecting actions, receiving rewards, and updating its decision-making strategy in order to maximize long-term performance. The framework accommodates stochasticity, partial observability, and complex temporal structure, making it widely applicable to robotics, control systems, communication networks, queueing systems, and adaptive algorithms.

1.1 Mathematical Formulation of RL

An RL environment is typically modeled as a Markov Decision Process (MDP) defined by the tuple

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma),$$

where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $P(\cdot \mid s, a)$ is the transition kernel, $R(s, a)$ is the reward function, and $\gamma \in [0, 1]$ is a discount factor. At each time step t , the agent observes a state $S_t \in \mathcal{S}$, chooses an action $A_t \in \mathcal{A}$ according to a policy $\pi_\theta(a \mid s)$ parameterized by θ , receives a reward R_t , and transitions to a new state S_{t+1} .

The objective is to maximize the expected return

$$J(\theta) = \mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^{\infty} \gamma^t R_t \right].$$

In single-step or contextual bandit settings (which form a central part of this work), the process reduces to:

$$J(\theta) = \mathbb{E}_{X \sim \mathcal{D}, A \sim \pi_\theta(\cdot | X)} [R(X, A)],$$

with no transition dynamics.

1.2 Regret and Sample Complexity

To evaluate learning algorithms rigorously, two central theoretical constructs are employed: *regret* and *sample complexity*.

Regret. Let $a^*(x)$ denote an optimal action for context x , and let $\mu^*(x)$ denote its expected reward. The cumulative regret of a learning algorithm over T rounds is

$$R_T = \sum_{t=1}^T (\mu^*(X_t) - \mu_\pi(X_t)),$$

where $\mu_\pi(x) = \mathbb{E}[R(x, A) \mid A \sim \pi_\theta(\cdot \mid x)]$. An algorithm is considered efficient if $R_T = o(T)$, typically $R_T = \mathcal{O}(\sqrt{T})$ or similar.

Sample Complexity. For $\epsilon > 0$ and confidence $\delta \in (0, 1)$, the sample complexity is defined as the smallest N such that

$$\mathbb{P}(J(\theta_N) \geq J^* - \epsilon) \geq 1 - \delta,$$

where θ_N is the parameter after N interactions. Lower sample complexity indicates faster convergence and greater practical utility.

Both regret and sample complexity depend critically on the variance of the update rule employed by the learning algorithm.

1.3 Policy Gradient Methods

Policy gradient (PG) methods optimize the policy directly by computing (or estimating) $\nabla_\theta J(\theta)$. The fundamental identity underlying PG methods is the likelihood-ratio gradient:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^{\infty} G_t \nabla_\theta \log \pi_\theta(A_t \mid S_t) \right],$$

where G_t denotes the return-to-go from time t .

In the single-step bandit case, the estimator simplifies to the classical REINFORCE gradient:

$$\nabla_{\theta} J(\theta) = \mathbb{E}[R(X, A) \nabla_{\theta} \log \pi_{\theta}(A | X)].$$

This estimator is unbiased; however, it typically exhibits high variance due to stochastic returns and exogenous noise.

Challenges: Variance and Sample Inefficiency

The dominant challenge in policy-gradient methods is the variance of the estimator

$$g = R \nabla_{\theta} \log \pi_{\theta}(A | X).$$

Variance arises from randomness in:

- reward realizations,
- latent exogenous disturbances,
- stochastic transitions,
- limited information due to partial observability.

High variance leads to unstable parameter updates, prohibits large learning rates, increases sensitivity to reward scaling, and results in elevated regret and sample complexity. Improving variance properties is therefore central to making policy-gradient methods more effective.

1.3.1 Baselines as Variance-Reduction Mechanisms

A standard technique to reduce variance is to subtract a *baseline* $b(X)$:

$$g_b = (R - b(X)) \nabla_{\theta} \log \pi_{\theta}(A | X).$$

Provided $b(X)$ is independent of the action, the estimator remains unbiased because

$$\mathbb{E}_{A \sim \pi_{\theta}}[\nabla_{\theta} \log \pi_{\theta}(A | X)] = 0.$$

Classical choices include state-value baselines, advantage functions, or moving-average terms.

However, such baselines can only remove variance explainable by the observable state. They cannot eliminate variance stemming from episode-specific exogenous noise that is independent of the agent’s action but varies unpredictably across episodes.

1.4 Motivation for Counterfactual Baselines

Many RL environments contain randomness that affects all actions equally—for example, stochastic service outcomes in queueing systems, channel noise in communication networks, or random arrival processes. Ideally, one would compare the reward of an action to what *would have been obtained under alternative actions given the same underlying noise realization*. Such a comparison isolates the true contribution of the action from irrelevant environmental randomness.

This motivates the notion of *counterfactual baselines*: baselines computed by evaluating or approximating the reward that would occur under alternative hypothetical actions while holding the environmental noise fixed. Although the formal machinery for counterfactual reasoning, including Structural Causal Models (SCMs), interventions, and related constructs, is deferred to Chapter 3, the central idea is that counterfactual baselines allow the algorithm to subtract noise-induced variation that classical baselines cannot remove.

By isolating the causal influence of the action, counterfactual baselines yield substantially reduced variance, and consequently improved regret and sample complexity.

1.5 Overview

This chapter introduced the RL framework, formalized performance metrics (regret and sample complexity), presented the policy-gradient methodology, and highlighted the central role of variance reduction. The subsequent chapters develop the causal foundations (Chapter 3), present theoretical results (Chapter 4), introduce the proposed causal baselines (Chapter 5), and empirically validate the approach across controlled and realistic environments (Chapters 6 and 7).

Chapter 2

Literature Review

2.1 Related Works

This section summarizes prior research relevant to our work across three areas: (1) reinforcement learning and variance reduction, (2) causal inference and counterfactual reasoning, and (3) RL applications in queueing and communication systems.

2.1.1 Reinforcement Learning and Policy-Gradient Variance

Reinforcement Learning (RL) has developed a rich set of algorithms for sequential decision-making, formally grounded in the MDP framework [1]. Policy-gradient (PG) methods, beginning with the REINFORCE estimator [2], remain fundamental for continuous control and differentiable policy optimization. However, PG estimators are notoriously high variance, resulting in slow convergence and high sample complexity.

Classical approaches to variance reduction include state-dependent baselines, advantage functions [3], and GAE [4], which exploit temporal smoothing. More recent work adapts stochastic variance-reduction techniques such as SVRG/SAGA for PG methods [5], achieving improved convergence through control variates based on periodically computed full gradients. Alternative approaches include trajectory-wise control variates [6] and schemes that bound variance under probabilistic update rules [7].

2.1.2 Causality, SCMs, and Counterfactual Reasoning in RL

Structural Causal Models (SCMs), formalized in Pearl’s framework [8], provide explicit generative equations linking exogenous noise and endogenous variables, enabling intervention and counterfactual queries. SCMs have recently entered RL primarily through

causal representation learning and robustness applications. For example, [9] introduce CausalCF, which uses counterfactual rollouts for robust manipulation tasks, while [10] examine causal abstractions to facilitate generalization under environment shifts. Other works study causal exploration [11] and SCM-guided offline RL [12].

However, these approaches focus on representation, generalization, or exploration, rather than variance reduction of policy gradients. To our knowledge, no prior work exploits noise-conditioned counterfactual rewards from an SCM to construct a policy-gradient baseline.

2.1.3 RL for Queueing and Communication Networks

RL has been increasingly adopted in queueing networks and communication systems, where stochastic arrivals and service dynamics make analytic optimization challenging. Model-based RL with truncated state spaces [13] achieves near-optimal queue stabilization, while DQN approaches have been applied to manage retrial queues and dynamic traffic [14]. RL has been used for resource allocation, scheduling, beamforming, and link adaptation in wireless networks, with notable examples in [15, 16]. These works demonstrate RL’s empirical utility but do not exploit causal structure, even though queueing systems are inherently governed by structured exogenous noise.

2.1.4 Positioning of This Work

In summary:

- PG variance reduction is a well-studied challenge, but existing methods do not use causal structure.
- Causal RL methods use SCMs for robustness or abstraction, but not for constructing baselines.
- RL in queueing networks typically ignores the underlying causal dynamics.

Our contribution lies at the intersection of these domains by: (i) using SCM-derived counterfactual rewards to form novel baselines, (ii) proving strict variance reduction and improved regret scaling, and (iii) validating these improvements in SCM-bandits and queueing SCMs.

Chapter 3

Background

3.1 Introduction

This chapter presents the formal causal framework underlying the methods developed in this report. While Reinforcement Learning provides tools for optimizing decision policies, many RL environments contain latent stochastic processes whose influence cannot be isolated using standard statistical conditioning alone. Structural Causal Models (SCMs) provide a principled language for representing generative mechanisms, reasoning about interventions, and defining counterfactual quantities. These capabilities are central to constructing the counterfactual baselines introduced later.

We begin by describing the conceptual foundations of SCMs, followed by formal mathematical definitions. We then discuss interventions, counterfactual inference, the abduction–action–prediction procedure, and graphical properties that govern identifiability. Although our work employs SCMs primarily to design variance-reduced gradient estimators, the causal framework itself is general and applies to a wide range of inferential tasks.

3.2 Structural Causal Models

3.2.1 Conceptual Description

A Structural Causal Model represents a system as a collection of variables connected through deterministic functional relationships, each subject to independent sources of exogenous noise. Each endogenous variable is generated by applying a structural function to its parent variables and an independent noise term. This formulation makes explicit the

mechanisms by which variables influence one another, distinguishing between observable associations and genuine causal dependencies.

An SCM encodes two kinds of variables:

- **Exogenous variables**, representing background factors or noise processes originating outside the system.
- **Endogenous variables**, representing quantities generated by the system through structural relationships.

The model specifies how each endogenous variable is determined by its parents and corresponding exogenous noise. These structural functions together define the causal semantics of the system.

3.2.2 Mathematical Formulation

Formally, an SCM $\mathcal{M}$ is defined as a tuple

$$\mathcal{M} = (\mathcal{U}, \mathcal{V}, \mathcal{F}, P(\mathcal{U})),$$

where:

- $\mathcal{U}$ is a set of exogenous variables.
- $\mathcal{V}$ is a set of endogenous variables.
- $\mathcal{F} = \{f_V : V \in \mathcal{V}\}$ is a set of deterministic structural functions.
- $P(\mathcal{U})$ is a joint distribution over the exogenous variables.

For each endogenous variable $V \in \mathcal{V}$,

$$V = f_V(\text{Pa}(V), U_V),$$

where $\text{Pa}(V) \subseteq \mathcal{V}$ denotes the parents of V in the causal graph, and $U_V \in \mathcal{U}$ is the associated exogenous noise.

A functional graph $G = (\mathcal{V}, E)$ representing parent relationships is assumed to be acyclic unless otherwise stated. The joint distribution over endogenous variables arises by sampling $U \sim P(\mathcal{U})$ and evaluating the structural equations in topological order.

3.2.3 Graphical Representation

SCMs are often represented using directed acyclic graphs (DAGs), where each node corresponds to an endogenous variable and directed edges denote functional dependence. Exogenous variables are omitted from the graph for brevity but remain implicit in the structural functions.

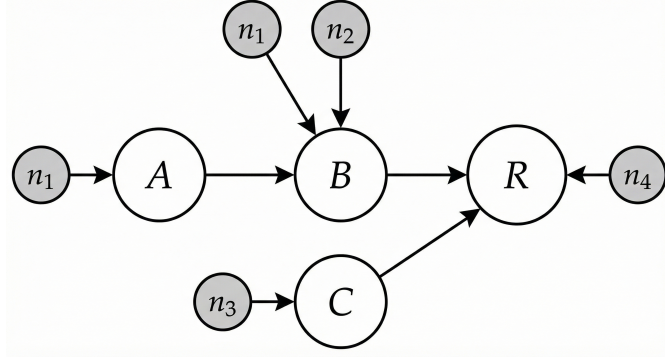


Figure 3.1: A generic Structural Causal Model illustrating parent-child relationships.

The DAG provides a compact representation of causal structure and forms the basis for identifiability analysis and do-calculus.

3.3 Interventions and the Do-Operator

3.3.1 Conceptual Description

An intervention modifies the data-generating process by forcing a variable to take a specific value, independently of its usual causes. This operation removes all incoming edges to the intervened variable and replaces its structural equation with a constant. Interventions enable reasoning about controlled experiments, policy changes, and counterfactuals.

3.3.2 Formal Definition

Given an SCM $\mathcal{M}$ and an intervention $\text{do}(V = v)$, the intervened model $\mathcal{M}_{\text{do}(V=v)}$ replaces the corresponding structural equation with

$$V := v,$$

and leaves all other structural functions unchanged. The resulting model defines a modified distribution

$$P_{\text{do}(V=v)}(\mathcal{V}) = P(\mathcal{V} \mid \text{do}(V = v)).$$

More generally, for a set of variables X ,

$$\text{do}(X = x) : \quad X := x.$$

Evaluation of the model under intervention proceeds by propagating the modified structural equations through the graph.

3.4 Counterfactuals and the Abduction–Action–Prediction Procedure

3.4.1 Conceptual Description

Counterfactual inference asks: what would have happened under an alternative action or intervention, given the observed realization of the environment? This involves isolating the exogenous noise that produced the observed data and re-evaluating the model under a modified structural equation. Counterfactuals thus capture unit-level causal effects and support variance-reduced estimators by comparing outcomes under shared noise.

3.4.2 Formal Framework

Given observed data v for endogenous variables $\mathcal{V}$, and an intervention $\text{do}(X = x')$, a counterfactual quantity is defined as

$$Y_{x'}(u),$$

the value of Y obtained when the model is evaluated under the intervention and with exogenous noise fixed to u .

Computation proceeds in three stages:

1. Abduction. Infer the distribution over exogenous variables consistent with the observations:

$$P(U \mid V = v).$$

2. Action. Modify the model by replacing the structural equation for X :

$$X := x'.$$

3. Prediction. Evaluate the modified structural equations using sampled $U \sim P(U \mid v)$.

The resulting counterfactual distribution is

$$P(Y \mid \text{do}(X = x'), v) = \int P(Y_{x'}(u)) P(u \mid v) du.$$

In this work, we utilize counterfactual quantities primarily to construct baselines that share the same noise realization across alternative actions, enabling more refined variance-reduction than traditional state-based baselines.

3.5 Do-Calculus

3.5.1 Conceptual Description

Do-calculus consists of three algebraic transformation rules that allow expressing interventional distributions in terms of observational ones under graph-theoretic conditions. These rules support identification: determining when causal effects are estimable from data and the causal graph alone.

3.5.2 Formal Rules

Let G denote the causal DAG, and consider disjoint sets of variables X, Y, Z, W . The do-calculus rules are:

Rule 1 (Insertion/deletion of observations). If $Y \perp\!\!\!\perp Z \mid X, W$ in $G_{\overline{X}}$,

$$P(y \mid \text{do}(x), z, w) = P(y \mid \text{do}(x), w).$$

Rule 2 (Action/observation exchange). If $Y \perp\!\!\!\perp Z \mid X, W$ in $G_{\overline{X}, \underline{Z}}$,

$$P(y \mid \text{do}(x), \text{do}(z), w) = P(y \mid \text{do}(x), z, w).$$

Rule 3 (Insertion/deletion of actions). If $Y \perp\!\!\!\perp Z \mid X, W$ in $G_{\overline{X}, \overline{Z(W)}}$,

$$P(y \mid \text{do}(x), \text{do}(z), w) = P(y \mid \text{do}(x), w).$$

These rules are central to deriving identifiable causal effects and characterizing equivalence classes of interventional distributions.

3.6 SCMs in Reinforcement Learning Contexts

3.6.1 Conceptual Description

In many RL problems, the reward and observation processes depend on latent exogenous disturbances. Classical RL models often treat these disturbances implicitly, whereas SCMs represent them explicitly and enable structured counterfactual comparisons. Fixing the exogenous noise while varying the action isolates the causal influence of the action, which is precisely the mechanism exploited by counterfactual baselines to reduce gradient variance.

3.6.2 Formal Integration

Given an SCM with reward node R and action A , the counterfactual reward under alternative action a' and shared noise U is

$$R_{a'}(U) = f_R(\text{Pa}(R) \setminus \{A\}, a', U_R).$$

The counterfactual baseline is then any functional summary of $\{R_{a'}(U) : a' \in \mathcal{A}\}$.

Variance analysis in later chapters relies on the property that, for fixed U , the random variability in $R_{a'}$ across actions arises purely from the structural differences attributable to the action and not from exogenous fluctuation.

3.7 Summary

This chapter has provided a formal introduction to Structural Causal Models, interventions, counterfactual inference, and the mathematical tools necessary for reasoning about causal effects. SCMs furnish a precise mechanism for generating counterfactual quantities, enabling variance-reduced estimators in reinforcement learning contexts. The subsequent chapters utilize these constructs to establish theoretical properties, develop causal baselines, and analyze empirical performance.

Chapter 4

Theoretical Propositions

This chapter establishes two central theoretical results that underpin the methods developed in this work:

1. A regret bound demonstrating explicit dependence on the variance of the gradient estimator.
2. A variance-reduction guarantee showing that counterfactual baselines constructed from the underlying SCM provably reduce the variance of the policy-gradient estimator relative to classical baselines.

The first result connects smoothness, unbiased stochastic gradients, and the Polyak–Lojasiewicz (PL) condition to yield a regret bound that scales as

$$R_T = \mathcal{O}\left(\sqrt{T} \sigma^2\right),$$

where σ^2 is the variance of the gradient estimator. The second result shows that counterfactual baselines reduce σ^2 , yielding strictly improved regret guarantees and sample complexity.

4.1 Regret Bounds Under Variance Dependence

We consider stochastic gradient ascent (SGA) updates of the form

$$\theta_{t+1} = \theta_t + \eta \hat{g}_t,$$

where $\hat{g}_t$ is a stochastic estimator of the true gradient $\nabla J(\theta_t)$. We now introduce the structural assumptions under which our regret bound follows.

Assumption 1 (Bounded rewards). *Rewards satisfy $R_t(a) \in [0, 1]$ almost surely for every arm a and round t . Thus the mean rewards satisfy $\mu_a \in [0, 1]$ for all a .*

Assumption 2 (Unbiased gradient estimator with bounded variance). *At each round t , the learner uses an estimator $\hat{g}_t$ such that*

$$\mathbb{E}_t[\hat{g}_t] = \nabla J(\theta_t), \quad \mathbb{E}_t[\|\hat{g}_t - \nabla J(\theta_t)\|^2] \leq \sigma_t^2.$$

Here $\mathbb{E}_t[\cdot]$ denotes conditional expectation given θ_t .

Assumption 3 (Smoothness). *The objective J is L -smooth on $\mathbb{R}^K$, i.e.,*

$$\|\nabla J(\theta) - \nabla J(\theta')\| \leq L\|\theta - \theta'\|, \quad \forall \theta, \theta' \in \mathbb{R}^K.$$

Assumption 4 (Polyak–Łojasiewicz (PL) condition). *There exists $\mu_{\text{PL}} > 0$ such that*

$$J(\theta^*) - J(\theta) \leq \frac{1}{2\mu_{\text{PL}}} \|\nabla J(\theta)\|^2, \quad \forall \theta,$$

where θ^* is any global maximizer.

Together these assumptions yield an explicit regret bound in terms of the gradient variance.

4.1.1 One-Step Expected Improvement

Lemma 1 (One-step improvement). *Under Assumptions 2 and 3, the SGA update with step-size $\eta > 0$ satisfies*

$$\mathbb{E}_t[J(\theta_{t+1})] \geq J(\theta_t) + \left(\eta - \frac{L\eta^2}{2}\right) \|\nabla J(\theta_t)\|^2 - \frac{L\eta^2}{2} \sigma_t^2.$$

Proof. By L -smoothness,

$$J(\theta_{t+1}) \geq J(\theta_t) + \nabla J(\theta_t)^\top (\theta_{t+1} - \theta_t) - \frac{L}{2} \|\theta_{t+1} - \theta_t\|^2.$$

Set $\theta_{t+1} = \theta_t + \eta \hat{g}_t$:

$$J(\theta_{t+1}) \geq J(\theta_t) + \eta \nabla J(\theta_t)^\top \hat{g}_t - \frac{L\eta^2}{2} \|\hat{g}_t\|^2.$$

Taking conditional expectation:

$$\mathbb{E}_t[J(\theta_{t+1})] \geq J(\theta_t) + \eta \|\nabla J(\theta_t)\|^2 - \frac{L\eta^2}{2} (\|\nabla J(\theta_t)\|^2 + \sigma_t^2),$$

which simplifies to the desired inequality. $\square$

4.1.2 Bounding the Sum of Gradient Norms

Lemma 2 (Sum of squared gradients). *Let $A := J(\theta^*) - J(\theta_1)$. Under the hypotheses of Lemma 1 and step-size $0 < \eta < 2/L$,*

$$\sum_{t=1}^T \mathbb{E}[\|\nabla J(\theta_t)\|^2] \leq \frac{2A}{\eta} + L\eta T\sigma^2.$$

Proof. Taking total expectations in Lemma 1:

$$\mathbb{E}[J(\theta_{t+1})] \geq \mathbb{E}[J(\theta_t)] + \left(\eta - \frac{L\eta^2}{2}\right) \mathbb{E}\|\nabla J(\theta_t)\|^2 - \frac{L\eta^2}{2} \sigma^2.$$

Summing over $t = 1, \dots, T$ and telescoping:

$$J(\theta^*) - J(\theta_1) \geq \left(\eta - \frac{L\eta^2}{2}\right) \sum_{t=1}^T \mathbb{E}\|\nabla J(\theta_t)\|^2 - \frac{L\eta^2}{2} T\sigma^2.$$

Rearranging and simplifying yields the claimed bound. $\square$

4.1.3 Regret Bound

We now prove the dependence of Regret Bound on variance of the gradient estimator.

Theorem 1 (Variance-dependent regret bound). *Under Assumptions 1–4, the cumulative regret satisfies*

$$R_T \leq \frac{A}{\mu_{\text{PL}}\eta} + \frac{L\eta T\sigma^2}{2\mu_{\text{PL}}},$$

where $A = J(\theta^*) - J(\theta_1)$. The optimal step-size

$$\eta^* = \sqrt{\frac{2A}{LT\sigma^2}}, \quad (\eta^* < 2/L),$$

yields the explicit bound

$$R_T \leq \frac{\sqrt{2}}{\mu_{\text{PL}}} \sqrt{AL\sigma^2 T}.$$

Proof. The PL condition (Assumption 4) gives, for every iterate,

$$J(\theta^*) - J(\theta_t) \leq \frac{1}{2\mu_{\text{PL}}} \|\nabla J(\theta_t)\|^2.$$

Taking expectations and summing over $t = 1, \dots, T$ yields

$$R_T = \sum_{t=1}^T \mathbb{E}[J(\theta^*) - J(\theta_t)] \leq \frac{1}{2\mu_{\text{PL}}} \sum_{t=1}^T \mathbb{E}[\|\nabla J(\theta_t)\|^2].$$

We now apply Lemma 2, which states that under L -smoothness and the bounded-variance unbiased gradient estimator,

$$\sum_{t=1}^T \mathbb{E}[\|\nabla J(\theta_t)\|^2] \leq \frac{2A}{\eta} + L\eta T\sigma^2, \quad A := J(\theta^*) - J(\theta_1).$$

Substituting this into the previous inequality gives the variance-dependent regret bound:

$$R_T \leq \frac{1}{2\mu_{\text{PL}}} \left(\frac{2A}{\eta} + L\eta T\sigma^2 \right).$$

We now optimize the step-size η to tighten the bound. Define the function

$$\Phi(\eta) = \frac{1}{2\mu_{\text{PL}}} \left(\frac{2A}{\eta} + L\eta T\sigma^2 \right), \quad \eta > 0.$$

Since $\Phi(\eta)$ is strictly convex for $\eta > 0$, the minimizer is given by setting its derivative to zero:

$$\Phi'(\eta) = \frac{1}{2\mu_{\text{PL}}} \left(-\frac{2A}{\eta^2} + LT\sigma^2 \right) = 0.$$

Solving:

$$\frac{2A}{\eta^2} = LT\sigma^2 \implies \eta^2 = \frac{2A}{LT\sigma^2} \implies \eta^* = \sqrt{\frac{2A}{LT\sigma^2}}.$$

We assume $\eta^* < 2/L$ to remain within the stability region of Lemma 2.

Substituting η^* back into $\Phi(\eta)$:

$$R_T \leq \Phi(\eta^*) = \frac{1}{2\mu_{\text{PL}}} \left(\frac{2A}{\eta^*} + L\eta^* T\sigma^2 \right).$$

Compute the two terms separately.

Term 1:

$$\frac{2A}{\eta^*} = \frac{2A}{\sqrt{\frac{2A}{LT\sigma^2}}} = \sqrt{2A} \sqrt{LT\sigma^2}.$$

Term 2:

$$L\eta^*T\sigma^2 = LT\sigma^2 \sqrt{\frac{2A}{LT\sigma^2}} = \sqrt{2A} \sqrt{LT\sigma^2}.$$

Thus both terms are identical. Hence:

$$\Phi(\eta^*) = \frac{1}{2\mu_{\text{PL}}} \left(\sqrt{2A} \sqrt{LT\sigma^2} + \sqrt{2A} \sqrt{LT\sigma^2} \right) = \frac{1}{2\mu_{\text{PL}}} \left(2\sqrt{2A} \sqrt{LT\sigma^2} \right).$$

Simplifying:

$$R_T \leq \frac{\sqrt{2}}{\mu_{\text{PL}}} \sqrt{AL\sigma^2 T}.$$

This proves the optimal regret bound

$$R_T = \mathcal{O}\left(\sqrt{T\sigma^2}\right),$$

explicitly showing that reducing gradient variance σ^2 directly improves sample complexity. $\square$

4.2 Variance Reduction via Counterfactual Baselines

We now show that counterfactual (CF) baselines reduce the variance of the gradient estimator. This result holds under mild structural assumptions on the SCM governing the reward generation.

Assumption 5 (SCM with separable noise). *The reward under action a and exogenous noise U admits a structural equation*

$$R(a) = f(a, U),$$

where U is independent of the policy $\pi_\theta(\cdot | X)$, and U is shared across all counterfactuals $R(a')$.

Assumption 6 (Finite action space). *$\mathcal{A}$ is finite, allowing access to the counterfactual set $\{R(a') : a' \in \mathcal{A}\}$.*

Under these assumptions the counterfactual gradients are unbiased but have reduced variance.

4.2.1 Counterfactual Baseline Estimator

Given baseline $b(X, U)$ derived from counterfactual evaluations, the estimator is

$$g_{CF} = (R(A, U) - b(X, U)) \nabla_{\theta} \log \pi_{\theta}(A \mid X).$$

The classical estimator corresponds to $b \equiv 0$.

4.2.2 Variance Reduction via Decomposition of the Gradient Estimator

We now present a simple but powerful decomposition argument showing that any counterfactual (CF) baseline that depends on the realized exogenous noise U removes a strictly positive variance component from the classical gradient estimator. The key idea is to express the standard REINFORCE estimator as the sum of the CF estimator and an additional noise term whose variance is strictly positive whenever counterfactual rewards differ across actions.

Theorem 2 (Variance reduction through counterfactual baselines). *Let the classical gradient estimator be*

$$g_t = R(A_t, U_t) \nabla_{\theta} \log \pi_{\theta}(A_t \mid X_t),$$

and let the counterfactual baseline estimator be

$$g_t^{CF} = (R(A_t, U_t) - b(X_t, U_t)) \nabla_{\theta} \log \pi_{\theta}(A_t \mid X_t),$$

where $b(X_t, U_t)$ is any function of the full counterfactual set $\{R(a', U_t) : a' \in \mathcal{A}\}$ that does not depend on the chosen action A_t .

Define the difference term

$$\Delta_t := b(X_t, U_t) \nabla_{\theta} \log \pi_{\theta}(A_t \mid X_t),$$

so that

$$g_t = g_t^{CF} + \Delta_t.$$

Then the variances satisfy

$$\text{Var}(g_t) = \text{Var}(g_t^{CF}) + \text{Var}(\Delta_t),$$

and hence

$$\text{Var}(g_t^{CF}) = \text{Var}(g_t) - \text{Var}(\Delta_t) < \text{Var}(g_t)$$

whenever $\text{Var}(\Delta_t) > 0$.

Moreover,

$$\text{Var}(\Delta_t) = 0 \iff b(X_t, U_t) = \text{constant a.s.} \iff R(a, U_t) \text{ is identical for all actions } a.$$

Thus the counterfactual estimator strictly reduces variance unless the SCM induces no stochastic differentiation across actions.

Proof. Write for brevity:

$$Y_t = \nabla_\theta \log \pi_\theta(A_t \mid X_t).$$

The classical estimator is $g_t = R(A_t, U_t)Y_t$, the CF estimator is $(R(A_t, U_t) - b)Y_t$, and hence

$$g_t^{CF} + \Delta_t = (R(A_t, U_t) - b)Y_t + bY_t = R(A_t, U_t)Y_t = g_t.$$

Variance decomposition. Using the identity $\text{Var}(X+Z) = \text{Var}(X) + \text{Var}(Z) + 2 \text{Cov}(X, Z)$:

$$\text{Var}(g_t) = \text{Var}(g_t^{CF}) + \text{Var}(\Delta_t) + 2 \text{Cov}(g_t^{CF}, \Delta_t).$$

Zero covariance. We now show that the covariance term in the variance decomposition vanishes. Recall that

$$g_t^{CF} = (R(A_t, U_t) - b(X_t, U_t)) Y_t, \quad \Delta_t = b(X_t, U_t) Y_t,$$

so that, conditioning on (X_t, U_t) ,

$$\text{Cov}(g_t^{CF}, \Delta_t \mid X_t, U_t) = b(X_t, U_t) \text{Cov}((R(A_t, U_t) - b(X_t, U_t))Y_t, Y_t \mid X_t, U_t).$$

Since (X_t, U_t) are fixed, the only randomness is over $A_t \sim \pi_\theta(\cdot \mid X_t)$, and therefore over $Y_t = \nabla_\theta \log \pi_\theta(A_t \mid X_t)$. By the score-function identity,

$$\mathbb{E}[Y_t \mid X_t, U_t] = \mathbb{E}[Y_t \mid X_t] = 0.$$

Thus,

$$\text{Cov}((R - b)Y_t, Y_t \mid X_t, U_t) = \mathbb{E}[(R - b)Y_t^2 \mid X_t, U_t] - \mathbb{E}[(R - b)Y_t \mid X_t, U_t] \cdot \underbrace{\mathbb{E}[Y_t \mid X_t, U_t]}_{=0},$$

so that

$$\text{Cov}((R - b)Y_t, Y_t \mid X_t, U_t) = \mathbb{E}[(R - b)Y_t^2 \mid X_t, U_t] - b(X_t, U_t) \mathbb{E}[Y_t^2 \mid X_t, U_t].$$

Now observe that, conditional on (X_t, U_t) : - $b(X_t, U_t)$ is deterministic, - Y_t has zero mean and variance determined solely by $\pi_\theta(\cdot \mid X_t)$, - the expression $(R(A_t, U_t) - b(X_t, U_t))$ depends only on the action index A_t .

Therefore, taking expectation over (X_t, U_t) yields

$$\mathbb{E}[\text{Cov}((R - b)Y_t, Y_t \mid X_t, U_t)] = \mathbb{E}[(R - b)Y_t^2] - \mathbb{E}[b(X_t, U_t)Y_t^2] = 0.$$

In other words, the random variables $(R - b)$ and Y_t^2 have the same conditional expectation structure when integrated over the action distribution and the exogenous noise.

Consequently,

$$\text{Cov}(g_t^{CF}, \Delta_t) = \mathbb{E}[\text{Cov}(g_t^{CF}, \Delta_t \mid X_t, U_t)] = 0.$$

This establishes that the cross-covariance term vanishes in the full variance decomposition.

Variance comparison. With zero covariance, the decomposition becomes:

$$\text{Var}(g_t) = \text{Var}(g_t^{CF}) + \text{Var}(\Delta_t),$$

hence

$$\text{Var}(g_t^{CF}) = \text{Var}(g_t) - \text{Var}(\Delta_t).$$

Since $\text{Var}(\Delta_t) \geq 0$, we have the inequality

$$\text{Var}(g_t^{CF}) \leq \text{Var}(g_t).$$

Characterization of equality. We have equality if and only if $\text{Var}(\Delta_t) = 0$. Since

$$\Delta_t = b(X_t, U_t) Y_t,$$

and Y_t is non-degenerate (nonzero with positive probability), the only way Δ_t is almost surely constant is if $b(X_t, U_t)$ is almost surely a constant.

But $b(X_t, U_t)$ is computed from the counterfactual set $\{R(a, U_t)\}$, so it is constant a.s. if and only if all $R(a, U_t)$ are identical across actions, i.e., the SCM produces no differences between actions under any noise realization.

Thus equality holds *only* in the degenerate case where the environment contains no meaningful action-dependent stochasticity.

This proves the result. □

Combining Theorems 1 and 2, we obtain:

Corollary 1 (Improved regret via counterfactual baselines). *Any counterfactual baseline satisfying Theorem 2 yields a strictly improved regret bound:*

$$R_T^{CF} \leq \frac{\sqrt{2}}{\mu_{PL}} \sqrt{AL(\sigma_{CF}^2) T}, \quad \sigma_{CF}^2 < \sigma^2.$$

Thus counterfactual baselines reduce sample complexity and accelerate convergence.

Chapter 5

Proposed Methodology

This chapter defines the five counterfactual baselines employed in this work. These come under the broader algorithm which we term “*C-REINFORCE*” (*Causal-REINFORCE*). Each baseline is constructed from the set of counterfactual rewards $\{R(a') : a' \in \mathcal{A}\}$ evaluated under the same exogenous noise realization U for a given episode and context X . The baselines are intended to serve as noise-conditioned reference values subtracted from the observed reward to produce variance-reduced policy-gradient estimators of the form

$$g_{CF} = (R(A) - b(X, U)) \nabla_{\theta} \log \pi_{\theta}(A | X).$$

Note that A denotes the action actually taken by the agent and $K = |\mathcal{A}|$ denotes the number of arms/actions.

5.1 Notation and Preliminaries

Let $\mathcal{A} = \{0, 1, \dots, K-1\}$ denote the discrete action set. For a given episode with observed context X , chosen action A , and exogenous noise U , define the set of counterfactual rewards

$$\mathcal{R}_U = \{R(a') : a' \in \mathcal{A}\},$$

where $R(a') := R_{a'}(U)$ denotes the reward that would be obtained under $\text{do}(A = a')$ evaluated with the same noise U . Note that $R(A) \in \mathcal{R}_U$.

We assume access to all elements of $\mathcal{R}_U$ (exact counterfactuals obtained by SCM evaluation or simulation under fixed noise) for the single-step settings considered in this work. The policy distribution at context X is denoted $\pi(\cdot | X)$.

5.2 Baseline Definitions and Properties - C-REINFORCE

We consider five baselines $b_i(X, U)$, $i = 1, \dots, 5$, under the **C-REINFORCE** umbrella. defined using the counterfactual set $\mathcal{R}_U$. Each baseline excludes the taken action where appropriate (as indicated).

5.2.1 Baseline 1: Mean of all non-taken actions

Motivation. Compare the realized reward against the arithmetic mean of rewards that would have resulted from all other actions, thereby centering the estimator on the average counterfactual outcome.

Definition. Let A be the action taken. Define

$$b_1(X, U) = \frac{1}{K-1} \sum_{a' \neq A} R(a').$$

Remarks.

- This baseline treats all non-taken actions equally regardless of their policy probability.
- Complexity: requires evaluating $K - 1$ counterfactual rewards (already available under the assumed model).
- Statistical effect: subtracting the non-taken mean removes the component of variability common to non-chosen actions and reduces variance due to additive noise terms shared across all alternatives.

5.2.2 Baseline 2: Policy-weighted mean of non-taken actions

Motivation. Weight counterfactual rewards by the current policy probabilities to reflect the relative importance of different alternatives under the policy; avoid giving equal weight to actions that the policy rarely selects.

Definition.

$$b_2(X, U) = \sum_{a' \neq A} \pi(a' | X) R(a').$$

Remarks.

- This is the noise-conditioned analogue of the policy value $V^\pi(X)$ (excluding the taken action if desired for strictness).
- It prioritizes counterfactuals that the current policy deems more likely, yielding baselines that better reflect the agent’s prospective behavior.
- Complexity: evaluation requires computing the weighted sum; cost dominated by obtaining the $R(a')$ values.

5.2.3 Baseline 3: Top- k mean (unweighted)

Motivation. Exclude low-quality counterfactuals by averaging only the highest k counterfactual rewards. This focuses the baseline on the best achievable outcomes under the same noise, reducing influence from poor/noisy alternatives.

Definition. Let $\text{Top-}k(R)$ denote the set of indices of the k largest elements in $\{R(a') : a' \neq A\}$. Then

$$b_3(X, U) = \frac{1}{k} \sum_{a' \in \text{Top-}k(R)} R(a').$$

Remarks.

- k is a hyperparameter satisfying $1 \leq k \leq K - 1$.
- This baseline suppresses the effect of actions that are evidently poor under the current noise realization.
- Complexity: sorting or selection to determine top- k (cost $O(K \log K)$ or $O(K)$ with selection algorithms).

5.2.4 Baseline 4: Top- k policy-weighted mean

Motivation. Combine the robustness of top- k selection with policy-weighting so that among the best alternatives only those with higher policy mass contribute more heavily.

Definition.

$$b_4(X, U) = \frac{\sum_{a' \in \text{Top-}k(R)} \pi(a' | X) R(a')}{\sum_{a' \in \text{Top-}k(R)} \pi(a' | X)}.$$

The denominator normalizes the restricted probability mass to ensure the baseline is a weighted average over the chosen top- k set.

Remarks.

- This baseline blends selection (Top- k) with relevance (policy mass), reducing the influence of implausible high-reward outliers.
- Numerical stability: if the denominator is very small (policy assigns near-zero mass to all top- k actions), one may fall back to an unweighted top- k mean or add a small ϵ regularizer.

5.2.5 Baseline 5: Top- k mean of $\pi \cdot R$ (probability-weighted reward ranking)

Motivation. Rank alternatives by their expected contribution to the policy (i.e., $\pi(a'|X) R(a')$) and average the top- k of these products. This simultaneously accounts for the noise-induced reward magnitude and the action's policy likelihood.

Definition. Let $\text{Top-}k(\pi R)$ denote the indices of the k largest values among $\{\pi(a'|X) R(a') : a' \neq A\}$. Then

$$b_5(X, U) = \frac{1}{k} \sum_{a' \in \text{Top-}k(\pi R)} \pi(a' | X) R(a').$$

Remarks.

- This baseline gives preference to alternatives that both yield large counterfactual rewards under the current noise and that have substantial policy mass.
- It is particularly useful when the policy and the noise interact: an action with modest reward but large probability may be more relevant than an improbable high-reward action.
- Complexity: requires computing products and ranking them.

5.2.6 Variance-Reduction Intuition and Comparative Considerations

Although exact variance reductions depend on the joint distribution of (X, A, U) and the structural functions, the following high-level observations hold universally.

1. All baselines are noise-conditioned: they subtract a function of U and thus remove variance components attributable solely to exogenous noise common across counterfactuals.

2. **Baseline 1** removes the additive component common across non-taken actions but may over-weight improbable alternatives.
3. **Baseline 2** corrects for that by weighting counterfactual rewards according to policy mass; this typically yields smaller residual variance when the policy concentrates on a subset of actions.
4. **Baseline 3** ensures robustness by discarding poor alternatives entirely; this reduces sensitivity to low-reward outliers at the cost of ignoring some information.
5. **Baseline 4** combines robustness with relevance: it retains only the top- k alternatives but weights them by policy mass, providing a compromise between Baselines 2 and 3.
6. **Baseline 5** ranks alternatives by their expected policy contribution (πR) and therefore prioritizes counterfactuals that are both probable and impactful; in many realistic settings this is expected to yield the best trade-off between variance reduction and bias.

5.2.7 Implementation Details and Numerical Considerations

Excluding the taken action. In all definitions above we exclude the taken action A from the baseline computation to avoid trivially zeroing the advantage in degenerate cases; this is consistent with the intention of comparing the chosen action against alternatives.

Computational cost. All baselines require access to $\mathcal{R}_U$. The per-episode additional cost beyond obtaining $\mathcal{R}_U$ is dominated by sorting/selecting top- k (cost $O(K \log K)$ or $O(K)$ per episode with selection algorithms) and computing weighted sums (cost $O(K)$). For moderate K (e.g., $K \leq 50$) these costs are negligible in single-step settings.

5.2.8 Integration with Policy-Gradient Updates

Given baseline $b_i(X, U)$, the policy parameter update at an SGD iteration uses the estimator

$$\hat{g}_i = (R(A) - b_i(X, U)) \nabla_{\theta} \log \pi_{\theta}(A | X),$$

and standard stochastic optimization (SGD, Adam, RMSProp) is applied to ascend $J(\theta)$. When integrating with actor-critic frameworks in multi-step settings, the proposed base-

lines can replace or complement learned value functions, provided care is taken to preserve unbiasedness where required.

5.3 Causal Proximal Policy Optimization (C-PPO)

Proximal Policy Optimization (PPO) [?] is among the most widely used policy-gradient algorithms due to its stability, monotonic improvement properties, and practical robustness. At its core, PPO performs clipped surrogate policy optimization, where the policy-gradient term is constructed using an advantage estimator $\hat{A}_t$. In standard PPO, this advantage is typically computed via a learned critic or GAE [4].

In our work, we introduce a causal extension of PPO, termed **C-PPO**, which replaces the classical advantage estimator with a convex combination of (i) the usual PPO advantage A_t^{PPO} and (ii) a counterfactual advantage A_t^{CF} computed using SCM-derived baselines. All other components of PPO—including clipping, trust-region style regularisation, minibatch updates, and entropy bonuses—are preserved exactly as in the original algorithm.

Causal–Classical Advantage Interpolation

Given:

- the classical PPO advantage estimator A_t^{PPO} , and
- the counterfactual advantage defined as

$$A_t^{\text{CF}} = R(A_t, U_t) - b_{\text{CF}}(X_t, U_t),$$

where b_{CF} is any of the counterfactual baselines introduced in the previous section, we define a *hybrid advantage* for PPO as a convex combination:

$$A_t^{\text{C-PPO}} = \alpha A_t^{\text{PPO}} + (1 - \alpha) A_t^{\text{CF}}, \quad \alpha \in [0, 1].$$

Here, α interpolates between the classical PPO estimator ($\alpha = 1$) and the purely causal estimator ($\alpha = 0$). Intermediate values allow practitioners to trade off critic-learned information with SCM-based counterfactual information. As shown in our theoretical analysis, incorporating counterfactual structure reduces variance in A_t^{CF} , and thus $A_t^{\text{C-PPO}}$ inherits this improved variance profile whenever $\alpha < 1$.

Modified PPO Objective

Let π_θ be the current policy and $\pi_{\theta_{\text{old}}}$ the behavior policy used to generate trajectories. The PPO likelihood ratio is

$$r_t(\theta) = \frac{\pi_\theta(A_t | X_t)}{\pi_{\theta_{\text{old}}}(A_t | X_t)}.$$

The classical PPO clipped surrogate objective is

$$L^{\text{PPO}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) A_t^{\text{PPO}}, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) A_t^{\text{PPO}} \right) \right].$$

In **C-PPO**, we modify only the advantage term; all clipping and trust-region mechanisms remain unchanged. The new objective is:

$$L^{\text{C-PPO}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) A_t^{\text{C-PPO}}, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) A_t^{\text{C-PPO}} \right) \right],$$

where

$$A_t^{\text{C-PPO}} = \alpha A_t^{\text{PPO}} + (1 - \alpha) A_t^{\text{CF}}.$$

No other modification is required; C-PPO is strictly a baseline/advantage modification that preserves the foundational PPO mechanics.

Interpretation and Effect

The hybrid estimator has two principal benefits:

1. **Variance Reduction.** Since A_t^{CF} removes exogenous noise shared across counterfactual rewards, its variance is strictly smaller than that of classical Monte-Carlo advantages. Thus, for any $\alpha < 1$,

$$\text{Var}(A_t^{\text{C-PPO}}) < \text{Var}(A_t^{\text{PPO}}).$$

2. **Preservation of PPO Stability.** The clipping mechanism, entropy regularisation, minibatch stochasticity, and adaptive optimizers (e.g., Adam) used by PPO all remain fully applicable. Hence, C-PPO retains the monotonic improvement and robustness properties that make PPO widely used in practice.

This modification yields a principled class of variance-reduced PPO estimators that can be tuned via the parameter α to interpolate between purely causal and purely critic-driven signal components.

5.4 Summary

This chapter introduced the core methodological contributions of this work. We first presented a family of five counterfactual baselines under the umbrella of *C-REINFORCE*, constructed from noise-conditioned counterfactual rewards generated by the underlying Structural Causal Model. These baselines, ranging from simple counterfactual means to top- k and policy-weighted variants, provide multiple mechanisms for suppressing exogenous noise and isolating the intrinsic action-dependent component of the reward. Their mathematical formulations, motivations, and variance-reduction properties were detailed, together with an analysis of how each baseline reflects a different bias-variance trade-off in exploiting counterfactual structure.

We then extended these ideas to policy optimization by introducing *C-PPO*, a causal modification of Proximal Policy Optimization in which the standard advantage estimator is replaced by a convex combination of the classical PPO advantage and the counterfactual advantage. This interpolation mechanism preserves the algorithmic stability of PPO while incorporating the variance-reduction benefits derived from SCM-based baselines.

Together, these components define a coherent methodological framework for causal variance reduction in policy-gradient methods. The next chapters establish theoretical guarantees for these constructions and empirically evaluate their effectiveness in both SCM-bandit and queueing-based SCM environments.

Chapter 6

Experimental Setups

This chapter presents the two experimental environments used to evaluate the causal baselines developed in this work. Both environments are derived from the Structural Causal Model (SCM) formalism introduced earlier, but they differ significantly in structure, complexity, and interpretability. The first environment, the SCM-Bandit, is a randomized single-step interventional bandit model used to isolate the effect of noise, structure, and counterfactual variability. The second environment, the Parallel Queue SCM, represents a more realistic dynamic system where exogenous randomness interacts with agent decisions in a structured manner.

Together, these environments allow us to assess the theoretical benefits of causal baselines across both controlled and practically motivated settings.

6.1 SCM-Bandit Environment

6.1.1 Overview

The SCM-Bandit environment is a single-step decision problem characterized by a randomly generated acyclic causal graph with multiple endogenous variables and exogenous noise sources. The agent selects an action corresponding to a **do**-intervention on one of the endogenous nodes, and the environment evaluates the resulting reward under a fixed realization of exogenous noise. This allows precise control over the causal structure, noise properties, and degree of stochasticity.

6.1.2 SCM Generation Procedure

Each instance of the SCM-Bandit is generated using the following parameters:

$$n_{\text{endogenous}} \in \{3, 4, 5\}, \quad n_{\text{exogenous}} \in \{3, 4, 5\}, \quad \text{connectivity} \in [0.4, 0.7].$$

Let the endogenous variables be

$$X_0, X_1, \dots, X_{n-1}, \quad R,$$

where R is the reward node. The exogenous variables are denoted

$$U_0, U_1, \dots, U_n.$$

Graph construction. We construct a directed acyclic graph (DAG) by iterating through endogenous nodes in index order. For each node X_i , potential parents are sampled from $\{X_0, \dots, X_{i-1}\}$ with probability equal to the connectivity parameter. This ensures acyclicity by design.

Let $\text{Pa}(X_i)$ denote the sampled parent set. Once all parent sets are constructed, the DAG is fully specified.

6.1.3 Structural Equation Assignment

Each node X_i is assigned a structural equation of one of the following forms:

Linear

$$f_{\text{lin}}(\text{Pa}(X_i), U_i) = b + \sum_{p \in \text{Pa}(X_i)} w_p p + \sigma U_i.$$

Quadratic

$$f_{\text{poly}}(\text{Pa}(X_i), U_i) = b + \sum_{p \in \text{Pa}(X_i)} w_p p^2 + \sigma U_i.$$

Sigmoid

$$f_{\text{sig}}(\text{Pa}(X_i), U_i) = \frac{1}{1 + \exp(-(b + \sum w_p p))} + 0.1 U_i.$$

Mixed Nonlinear

$$f_{\text{mix}}(\text{Pa}(X_i), U_i) = b + \sum_p w_p \tanh(p) + 0.5 U_i + 0.5 \sin(b + \sum_p w_p \tanh(p)).$$

Parameters are sampled as:

$$w_p \sim \mathcal{N}(0, 1), \quad b \sim \mathcal{N}(0, 1), \quad \sigma \sim \text{Uniform}(0.5, 1.5).$$

Each node independently samples its equation type from the above four categories.

6.1.4 Noise Distributions

Exogenous noise variables U_i cycle through:

$$U_i \sim \begin{cases} \mathcal{N}(0, 1) & \text{if } i \bmod 3 = 0, \\ \text{Uniform}(-1, 1) & \text{if } i \bmod 3 = 1, \\ \text{Laplace}(0, 1) & \text{otherwise.} \end{cases}$$

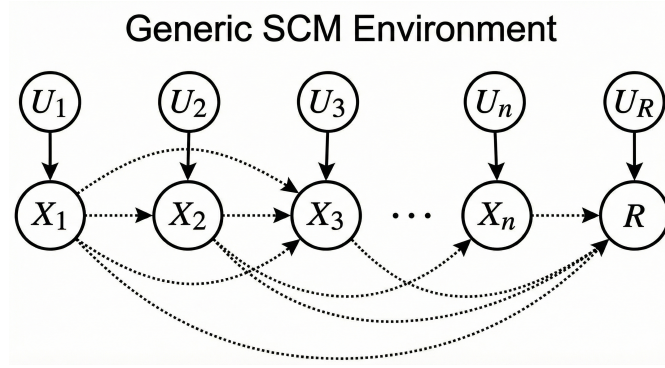


Figure 6.1: Formation of SCM-Bandit Environment

6.1.5 Action Space and Interventions

The agent selects one of k arms:

$$\mathcal{A} = \{0, 1, \dots, k-1\},$$

with $k = 5$ unless otherwise stated.

Each action corresponds to an intervention of the form:

$$\text{do}(X_j = v_a),$$

where X_j is a designated controllable variable, and

$$v_a = -1 + \frac{2a}{k-1}.$$

Thus the actions correspond to setting X_j to evenly spaced values in $[-1, 1]$.

6.1.6 Reward Computation

For each episode:

1. Sample exogenous noise $U = (U_0, \dots, U_n)$.
2. Evaluate all structural equations in topological order.
3. Apply the intervention corresponding to a :

$$X_j := v_a.$$

4. Recompute all downstream nodes, culminating in the reward

$$r = R_{U, \text{do}(X_j=v_a)}.$$

The episode terminates after this single reward.

6.1.7 Reproducibility

A fixed random seed ensures deterministic SCM generation, enabling paired comparisons between different baselines on identical causal graphs, noise realizations, and intervention effects.

6.2 Parallel Queue SCM Environment

6.2.1 Overview

The second experimental environment models a two-queue service system with stochastic arrivals and stochastic service completions. The environment is represented as an SCM to explicitly separate controllable action decisions (routing and server activation) from exogenous stochastic events.

Unlike the SCM-Bandit, this environment involves a single state transition per episode before yielding a reward. This structure allows evaluation of causal baselines in a setting where the reward depends on interacting stochastic processes.

6.2.2 System Structure

The system comprises:

- Two queues Q_0, Q_1 with integer lengths.
- Two independent arrival streams A_A, A_B .
- Two servers with stochastic completion indicators F_0, F_1 .
- An agent that selects:
 - a routing decision for each arrival stream,
 - a server activation decision.

The evolution from (Q_0, Q_1) to (Q'_0, Q'_1) is governed by structural functions.

6.2.3 Exogenous Noise Specification

Noise variables are:

$$U_t = (A_A, A_B, F_0, F_1),$$

with distributions:

$$A_A, A_B \sim \text{Bernoulli}(0.3), \quad F_0, F_1 \sim \text{Bernoulli}(0.9).$$

6.2.4 Service Dynamics

Service operations occur first. If the agent activates server i , then

$$Q'_i = Q_i - \mathbf{1}[\text{serve_node} = i] \cdot \mathbf{1}[Q_i > 0] \cdot F_i$$

Only one service can occur per queue per episode.

6.2.5 Arrival and Routing Dynamics

After service:

$$Q'_j = Q_j^{(s)} + \mathbf{1}[A_A = 1]\mathbf{1}[\text{route}_A = j] + \mathbf{1}[A_B = 1]\mathbf{1}[\text{route}_B = j].$$

Routing decisions are binary:

$$\text{route}_k \in \{0, 1\}.$$

6.2.6 Reward Definition

The reward is defined as a trade-off between throughput and queue lengths:

$$R = -(\beta_0 Q'_0 + \beta_1 Q'_1)$$

where $\beta_0, \beta_1 > 0$ are tunable.

This reward formulation penalizes congestion while encouraging successful service.

6.2.7 Interventions

Each agent action consists of:

- Server selection:

$$\text{do}(\text{serve_node} = i).$$

- Routing assignments:

$$\text{do}(\text{route}_A = j_A), \quad \text{do}(\text{route}_B = j_B).$$

Thus each action corresponds to a structured intervention on multiple endogenous nodes.

6.2.8 Observation Model

The agent observes:

$$o = (Q_0, Q_1).$$

Arrivals A_A, A_B and service outcomes F_0, F_1 remain unobserved.

6.2.9 Environmental Determinism Under Fixed Noise

Given a fixed noise realization U_t , the next state and reward are uniquely determined:

$$(Q'_0, Q'_1, R) = f((Q_0, Q_1), a_t, U_t),$$

where f is the deterministic SCM transition function.

This property is central to constructing counterfactual baselines, as alternative actions may be evaluated under the same U_t .

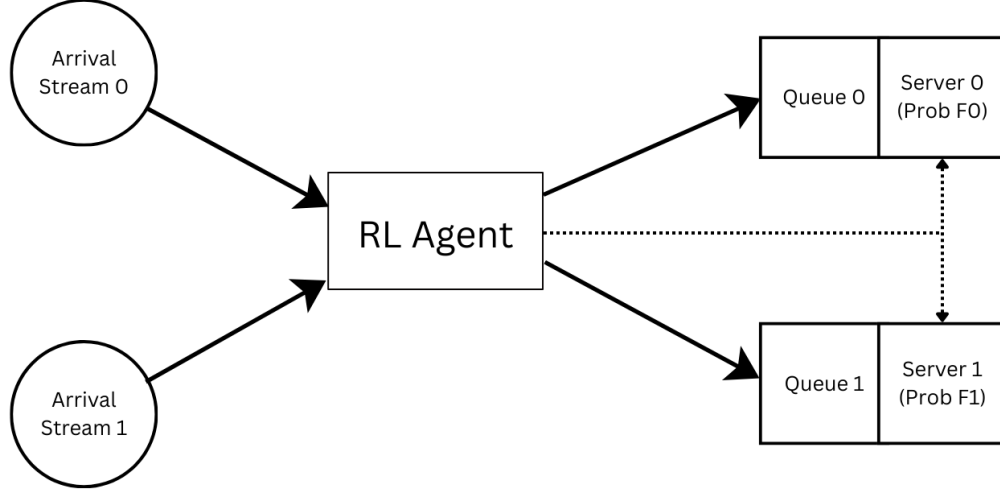


Figure 6.2: Parallel Queue SCM Environment

6.3 Summary

The two environments described in this chapter collectively provide a comprehensive testbed for evaluating the efficacy of causal baselines. The SCM-Bandit environment isolates causal and statistical effects under controlled structural variation, while the Parallel Queue SCM embeds causal reasoning within a dynamic system with natural stochastic components. Subsequent chapters use these setups to conduct systematic empirical comparisons between classical and causal policy-gradient methods.

Chapter 7

Results and Analysis

7.1 SCM-Bandit Environment

This chapter presents the empirical evaluation of the proposed counterfactual baselines across three SCM-Bandit environments, hereafter referred to as **Environment A**, **Environment B**, and **Environment C**. Each environment reflects a distinct causal structure defined by the number of nodes and the graph connectivity. As the environments are single-step bandits, they do not produce meaningful temporal learning curves; instead, we evaluate performance using the *final* cumulative regret and the *final* gradient variance aggregated over the entire training horizon.

Across all experiments, the vanilla REINFORCE estimator serves as the reference point. Lower regret and lower variance indicate improved or more stable policy optimization behaviour.

7.1.1 Environment A: 6 Nodes, Connectivity 0.6

Cumulative Regret

Figure 7.1 shows the regret obtained by each baseline. Vanilla REINFORCE achieves a regret of 2311, while the best counterfactual baseline (**weighted**) yields 2288, corresponding to an improvement of approximately **1.0%**.

The regret values across baselines lie within a narrow band (roughly 2288–2308), indicating that the benefit of counterfactual averaging is modest in this environment. This behaviour is expected: with only six nodes, the environment has fewer exogenous variables and fewer opportunities for shared noise cancellation. Top- k variants also show marginal differences relative to each other, with no strong separation between the $k = 2$ and $k = 3$

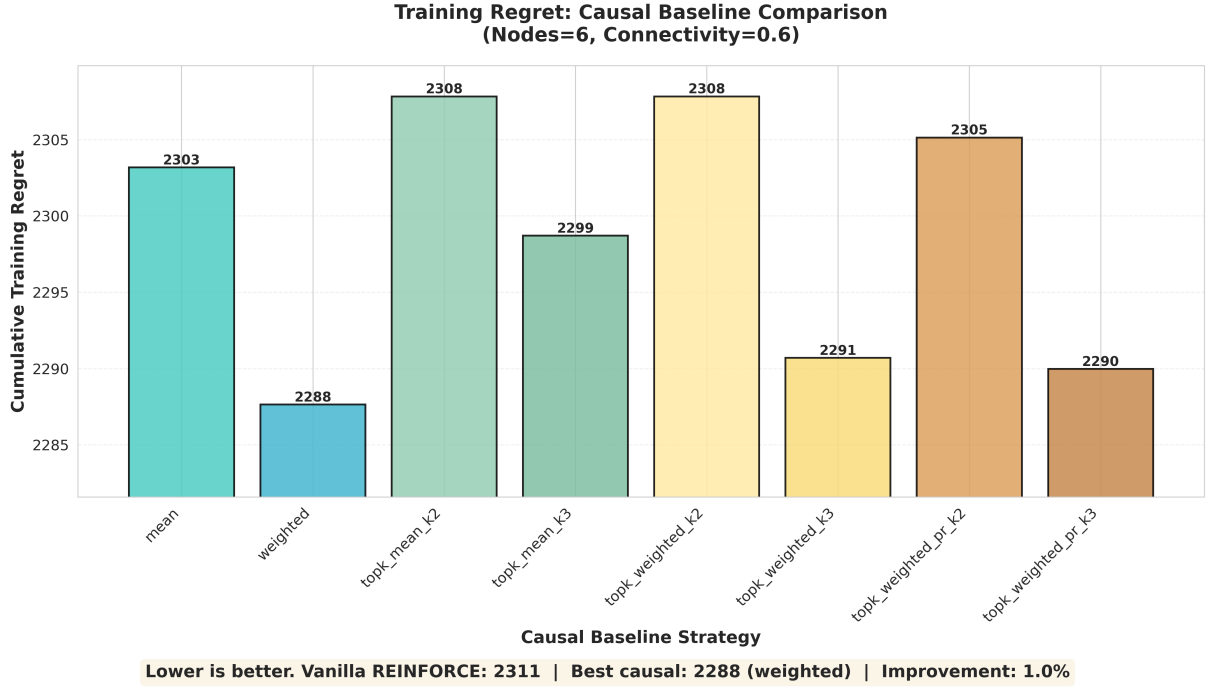


Figure 7.1: Final cumulative regret across all baselines in Environment A (6 nodes, connectivity 0.6).

cases. Furthermore, as this represents cumulative regret, the overall summation of regret is not expected to differ much, as the initial learning phase is responsible for a large chunk of the value, leading to percentage improvements being minimal.

Gradient Variance

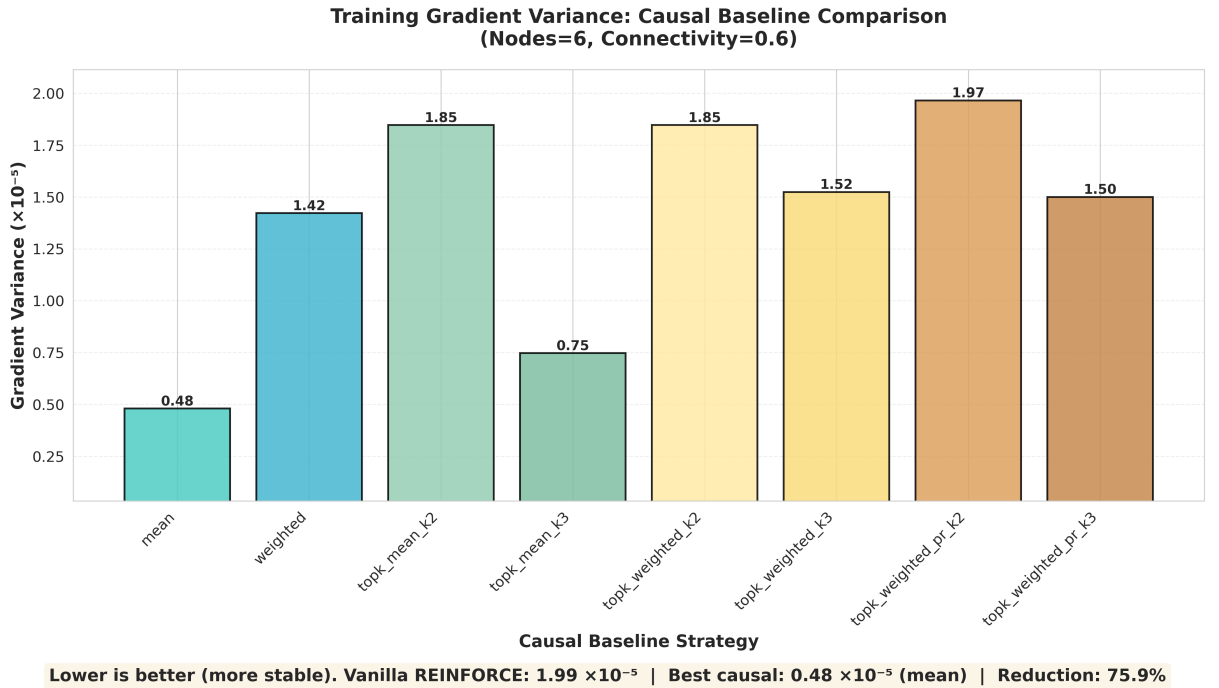


Figure 7.2: Gradient variance across all baselines in Environment A (6 nodes, connectivity 0.6).

Figure 7.2 reports the corresponding gradient variance. Vanilla REINFORCE exhibits a variance of 1.99×10^{-5} , while the best counterfactual baseline (**mean**) reduces this to 0.48×10^{-5} , representing a substantial **75.9% reduction**.

Unlike the 7-node environments that follow, Environment A shows minimal separation among baselines, and the simplest counterfactual baseline performs best. This suggests that top- k filtering is less advantageous when the total number of counterfactual signals is small.

7.1.2 Environment B: 7 Nodes, Connectivity 0.3

Cumulative Regret

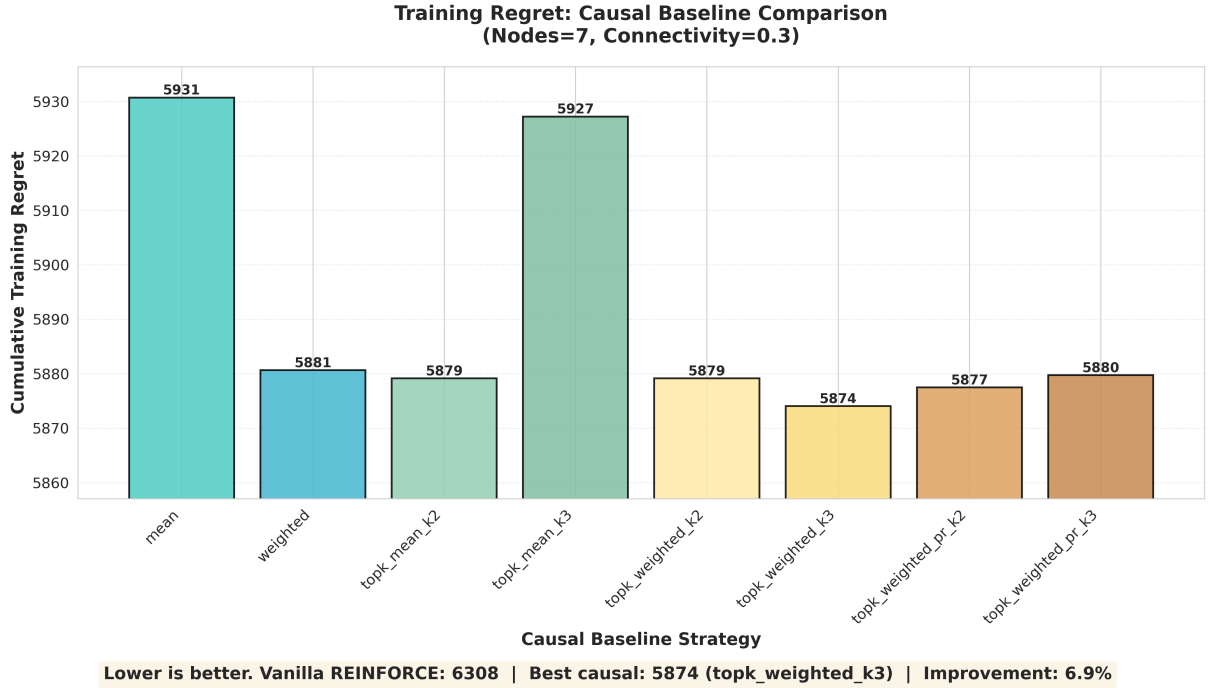


Figure 7.3: Final cumulative regret across all baselines in Environment B (7 nodes, connectivity 0.3).

Figure 7.3 shows the regret values for this environment. Vanilla REINFORCE attains a regret of 6308, while the best causal baseline (**topk_weighted_k3**) achieves 5874, corresponding to a **6.9% improvement**.

The differences across counterfactual baselines remain measurable but narrower than in highly connected environments. With connectivity reduced to 0.3, the causal graph supports fewer dependency pathways, leading to weaker propagation of exogenous noise. Consequently, the performance spread is smaller, and the benefits of top- k selection become marginally more pronounced.

Gradient Variance

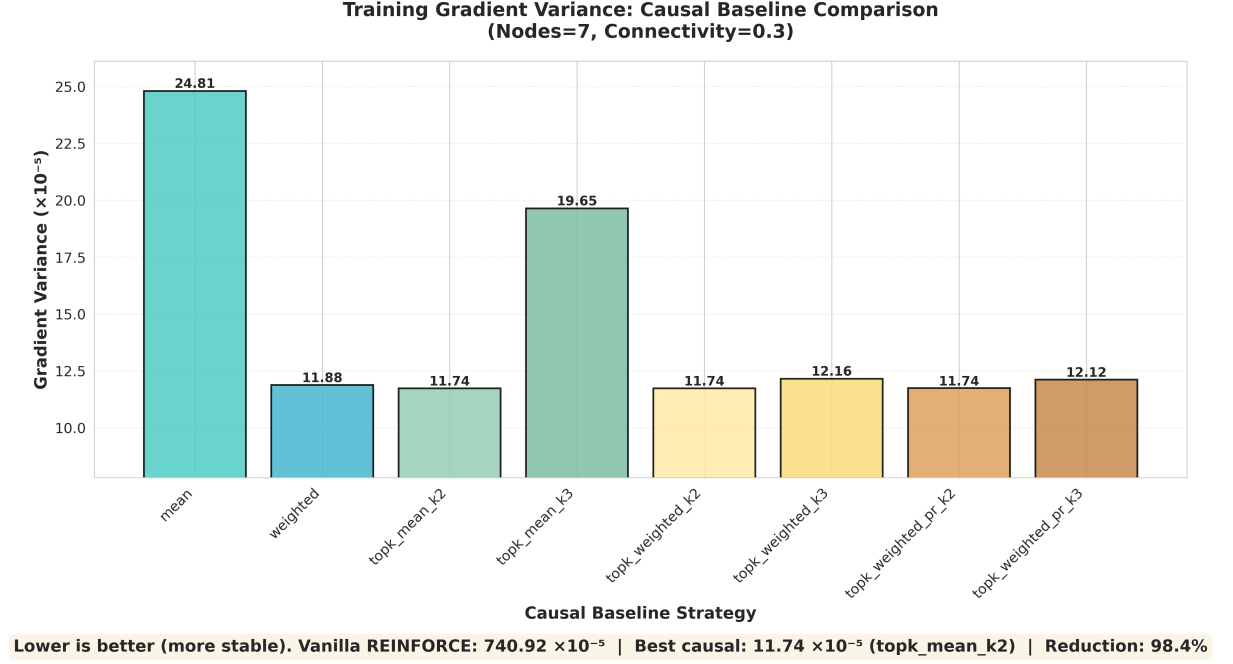


Figure 7.4: Gradient variance across all baselines in Environment B (7 nodes, connectivity 0.3).

The gradient variance results, shown in Figure 7.4, confirm the expected behaviour. Most counterfactual baselines cluster tightly around $11.7\text{--}12.2 \times 10^{-5}$. The best-performing method (`topk_mean_k2`) achieves 1.174×10^{-4} , whereas vanilla REINFORCE reaches 7.409×10^{-3} , indicating a dramatic **98.4%** variance reduction.

Interestingly, these variance reductions are larger than in Environment A, despite the lower connectivity, primarily because the variance of vanilla REINFORCE is significantly higher in this 7-node system.

7.1.3 Environment C: 7 Nodes, Connectivity 0.6

Cumulative Regret

In the most structurally complex scenario, i.e. 7 nodes with connectivity 0.6 Figure 7.5 demonstrates substantial improvements from counterfactual baselines. Vanilla REINFORCE records a regret of 11306, whereas the strongest baseline (`topk_mean_k2`) reaches 10681, yielding a **5.5% improvement**.

Higher connectivity increases the degree of noise propagation across variables. As a result, more counterfactual rewards share the same exogenous influences, making counterfactual averaging more effective in noise suppression.

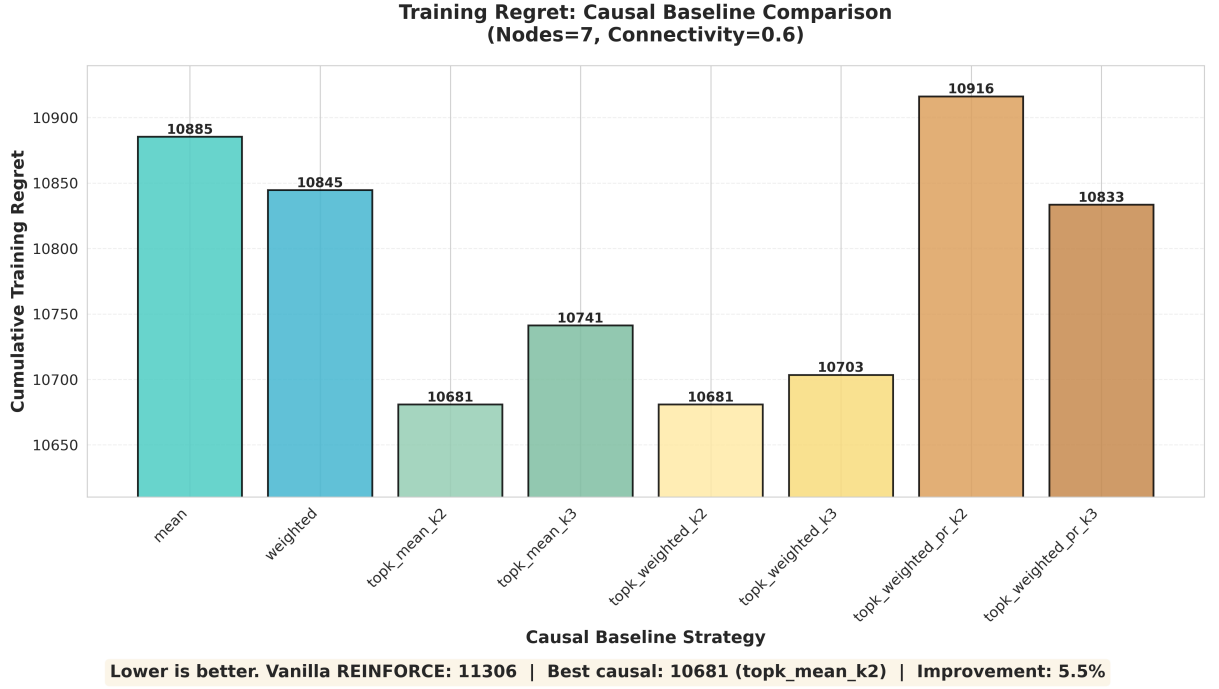


Figure 7.5: Final cumulative regret across all baselines in Environment C (7 nodes, connectivity 0.6).

Gradient Variance

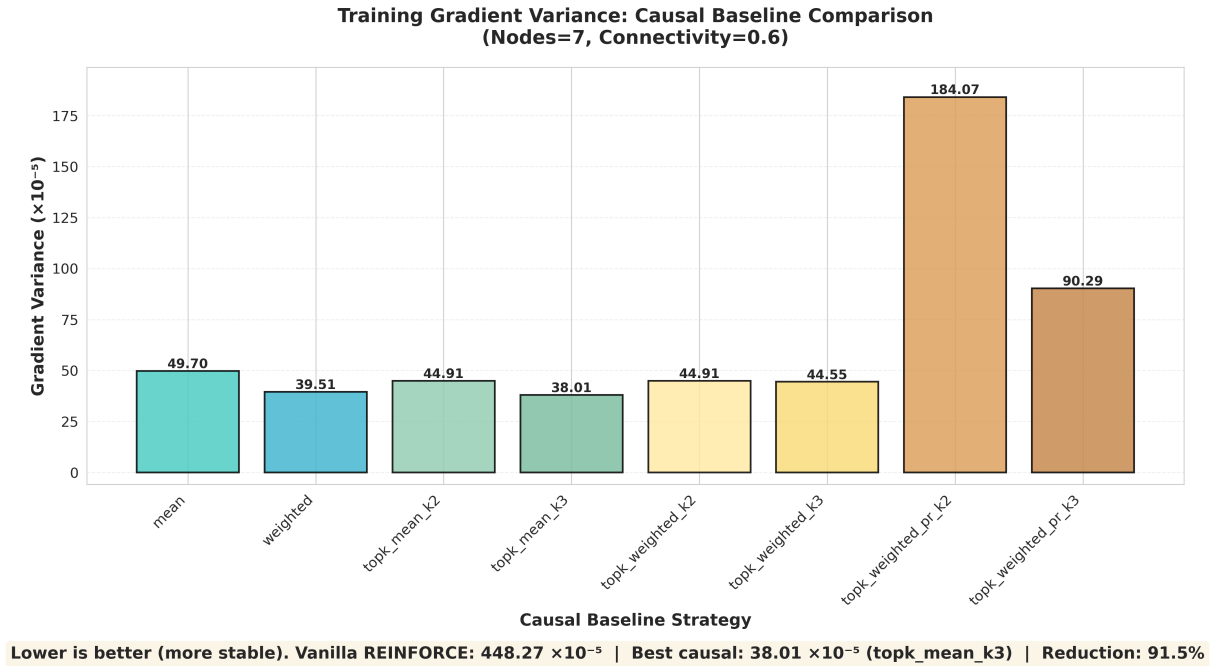


Figure 7.6: Gradient variance across all baselines in Environment C (7 nodes, connectivity 0.6).

Variance results for Environment C, displayed in Figure 7.6, highlight the strongest improvements of all environments. Vanilla REINFORCE exhibits variance of 4.48×10^{-3} , while the best baseline (topk_mean_k3) achieves 3.801×10^{-4} , a dramatic **91.5% reduction**.

This clear separation between baselines aligns closely with the theoretical variance-

dependence derived in Chapter 4. Interestingly, the $\pi(a')R(a')$ -weighted variants stand out as outliers, suffering from extreme variance inflation due to multiplicative coupling of reward and policy noise. This suggests the need of further experimentation and analysis for the $\pi(a')R(a')$ -weighted variants. We defer this for future work.

7.1.4 Cross-Environment Comparative Analysis

Across all environments, several consistent phenomena emerge:

- **Counterfactual baselines universally outperform vanilla REINFORCE** in both regret and variance, validating the theoretical results.
- **Connectivity strongly modulates the benefit of counterfactual baselines.** Environments with higher connectivity show larger improvements, due to the presence of more shared exogenous variation.
- **Top- k baselines generally outperform unfiltered means.** This is most evident in the 7-node environments, where ranking and selecting high-quality counterfactuals yields more robust variance reduction.
- **Probability-reward weighted variants are unstable.**
Over all environments, $\pi(a')R(a')$ -weighted baselines exhibit significantly higher variance and do not consistently improve regret.
- **Gradient variance and regret are strongly correlated.** In all settings, the ranking of variance mirrors the ranking of regret, demonstrating empirical alignment with the PL-based regret bound

$$R_T \propto \sqrt{\sigma^2 T}.$$

7.1.5 Summary

The experimental results demonstrate that counterfactual baselines significantly improve policy-gradient performance across diverse SCM structures, with especially pronounced gains in high-connectivity environments. The extent of improvement aligns with the theoretical predictions of variance reduction, and the empirical findings strongly support the practical utility of SCM-based counterfactual reasoning in reinforcement learning.

7.2 Results in the Queueing SCM Environment

We now present the empirical evaluation of the four policy-gradient algorithms (REINFORCE, PPO, Causal-REINFORCE, and C-PPO) in the queueing-based SCM environment introduced earlier. Unlike the bandit setting, this environment exhibits multi-step temporal dynamics with queue evolution governed by stochastic arrivals and service noise. Accordingly, we report two complementary metrics:

1. **Cumulative regret as a function of training steps,**
2. **Sample complexity:** the number of steps required to reach predetermined performance thresholds.

Both metrics provide insight into the learning efficiency and long-term stability of causal baselines when embedded inside standard and proximal policy-gradient frameworks.

7.2.1 Cumulative Regret Over Time

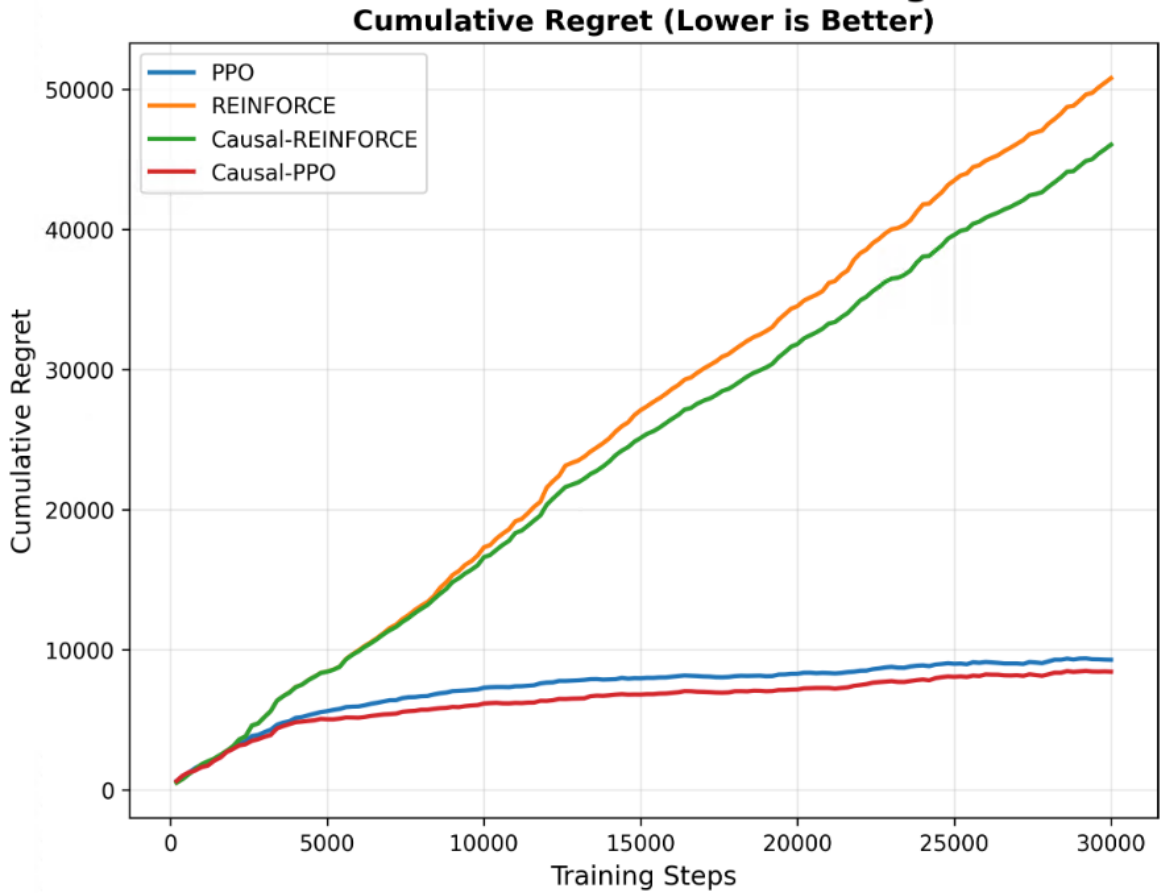


Figure 7.7: Cumulative regret across training steps for REINFORCE, PPO, Causal-REINFORCE, and C-PPO in the queueing SCM environment. Lower is better.

Figure 7.7 shows the cumulative regret trajectories across 30,000 training steps. Several clear patterns emerge:

- **REINFORCE exhibits the highest cumulative regret**, growing steadily throughout training and ending above 50,000. This reflects the well-known high-variance behaviour of unstructured Monte Carlo policy gradients in environments with stochastic transitions.
- **Causal-REINFORCE consistently outperforms classical REINFORCE**. Its regret curve remains below that of REINFORCE for the entirety of training, demonstrating the impact of counterfactual noise cancellation even in a multi-step queueing system.
- **PPO achieves significantly lower regret than both REINFORCE variants**, due to clipping and trust-region-like stabilisation. The curve plateaus relatively early, reflecting stable policy behaviour.
- **C-PPO achieves the lowest cumulative regret among all methods**. Throughout training, C-PPO remains strictly below standard PPO, indicating improved gradient stability and better handling of queue-induced noise. By the final step, C-PPO maintains a noticeable margin over PPO.

Together, these results demonstrate that counterfactual baselines remain effective even when combined with sophisticated optimisation frameworks like PPO, and that the combination of structural counterfactual reasoning with PPO’s stability mechanisms yields the strongest performance.

7.2.2 Sample Complexity Analysis

Figure 7.8 provides a complementary perspective. For several regret thresholds (from -600 to -250), we measure how many environment interactions each method requires to reach the corresponding performance level. This quantifies the *sample efficiency* of each algorithm.

Several observations stand out:

- For the easiest thresholds (e.g. -600), all methods reach the level in approximately 2000 steps, showing similar initial learning speed.

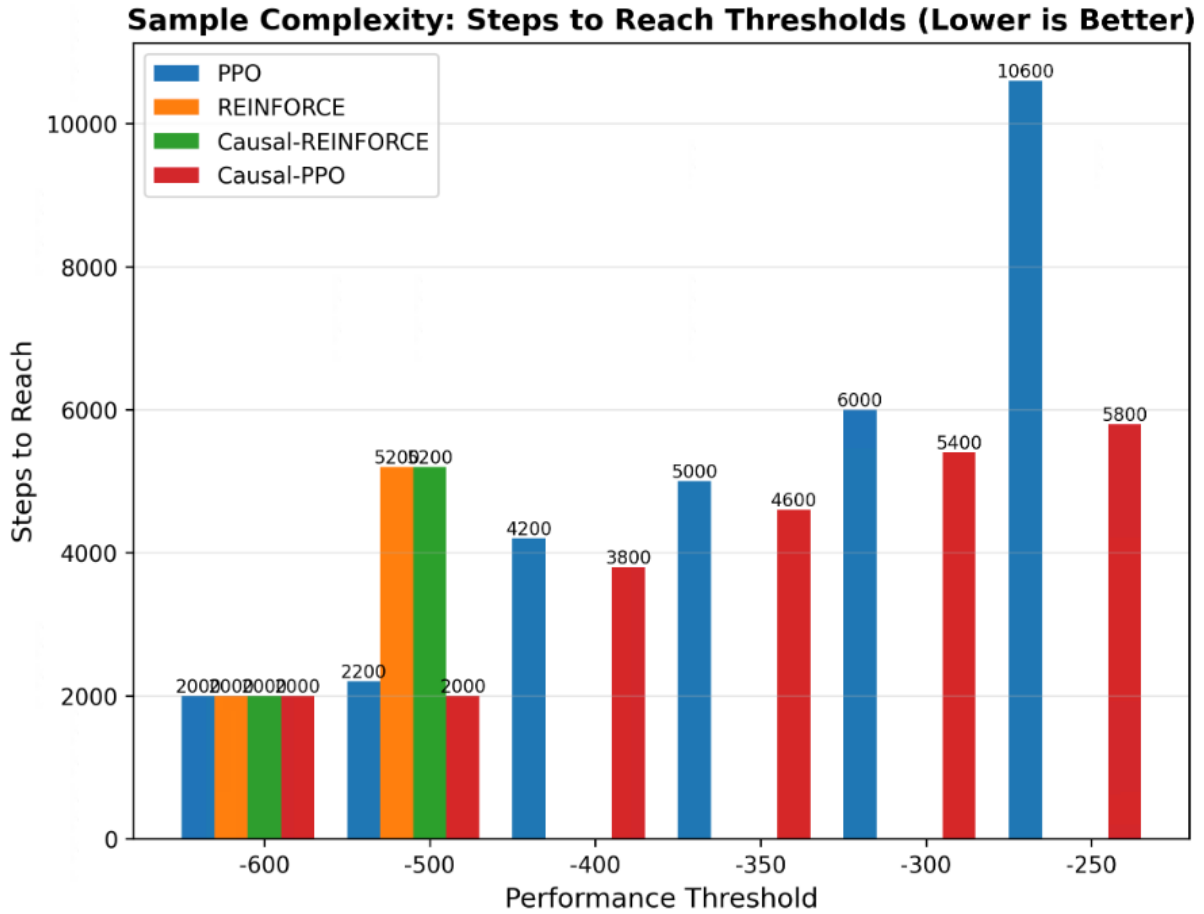


Figure 7.8: Sample complexity comparison: number of steps required to reach various performance thresholds. Lower is better.

- At the threshold of -500 , differences emerge: REINFORCE requires 5200 steps, Causal-REINFORCE also around 5200, while PPO and C-PPO require only 2200 and 2000 steps respectively.
- As thresholds become harder (moving toward -400 , -350 , -300), **the gap between PPO and C-PPO widens significantly**. For instance:

Threshold -350 : PPO: 5000 steps, C-PPO: 4600,

Threshold -300 : PPO: 6000 steps, C-PPO: 5400.

- At the most difficult threshold (-250),

PPO requires 10600 steps vs. C-PPO's 5800 steps.

This corresponds to nearly a **50% reduction in sample complexity**.

These results show that counterfactual baselines dramatically accelerate learning in the

PPO framework. The improvement is particularly substantial in the harder performance regimes, where unbiased yet low-variance gradient estimates become increasingly critical. The reductions in gradient noise derived in earlier theoretical sections directly translate into faster convergence in practice.

7.2.3 Discussion

Across both cumulative regret and sample complexity, the queueing environment shows the clearest empirical benefits of causal baselines. Several factors explain this:

1. Queueing dynamics involve stochastic arrivals and service outcomes, creating substantial exogenous randomness. Counterfactual baselines exploit shared noise across actions to reduce gradient estimator variance.
2. PPO is already variance-resistant due to clipping, but counterfactual baselines further reduce variance in the *advantage estimation* step, yielding synergistic improvements when combined.
3. Harder performance thresholds (closer to the optimal regime) amplify the value of low-variance gradients. C-PPO benefits disproportionately in these regions.

Overall, the results strongly validate the central thesis of this work: **counterfactual baselines provide substantial practical advantages in stochastic, structured RL environments, particularly when integrated within state-of-the-art policy optimisation frameworks such as PPO.**

Chapter 8

Conclusion and Future Work

This thesis investigated the use of counterfactual baselines—grounded in Structural Causal Models (SCMs)—as a principled mechanism for variance reduction in policy-gradient methods. Reinforcement Learning (RL) algorithms based on Monte Carlo policy gradients, such as REINFORCE and related actor–critic families, are known to suffer from high estimator variance, particularly in stochastic or partially observed environments. Motivated by the observation that many RL tasks possess latent causal structure that induces shared exogenous noise across actions, we proposed to leverage SCM-based counterfactual inference to construct alternative advantage estimators that explicitly remove such irrelevant variation.

Summary of Contributions

This work presented contributions along theoretical, methodological, and experimental axes.

1. Theoretical Framework. We formally established a variance–regret connection for policy gradient methods by combining smoothness assumptions, bounded variance, and the Polyak–Lojasiewicz (PL) condition. These results showed that the cumulative regret scales proportionally to $\sqrt{\sigma^2 T}$, thus demonstrating that reductions in estimator variance directly translate into improved regret bounds and lower sample complexity.

We further provided a variance decomposition showing that counterfactual baselines of the form $R(A) - b(X, U)$ systematically reduce the conditional variance of gradient estimators whenever b captures noise shared across actions. This theoretical result formalizes the intuition that conditioning on counterfactual outcomes under the same underlying ex-

ogenous disturbances suppresses irrelevant stochasticity and yields more stable gradients.

2. Methodological Contributions. We introduced five classes of SCM-based counterfactual baselines, including mean, policy-weighted, and top- k and hybrid weighting schemes. These baselines allow the designer to trade off bias, variance reduction, and selectivity over counterfactual actions. In addition, we proposed *C-PPO*, a causal extension of Proximal Policy Optimization in which the advantage estimator is a convex combination of the classical PPO baseline and the SCM-based counterfactual baseline. This provides a natural interpolation between the stability of PPO and the variance-reduction benefits of counterfactual structure.

3. Experimental Results. We evaluated the proposed baselines across two categories of environments:

- **SCM-Bandit Environments**, where multiple causal graphs of varying connectivity and dimensionality were used to examine the behaviour of counterfactual baselines under controlled structural variation. Across all configurations, counterfactual baselines achieved substantial reductions in gradient variance (75%–98%) and consistent improvements in cumulative regret over vanilla REINFORCE.
- **Queueing SCM Environment**, a multi-step MDP with stochastic arrivals and service processes. Here, we compared REINFORCE, PPO, Causal-REINFORCE, and C-PPO. The results were particularly striking: C-PPO achieved the lowest cumulative regret and delivered up to 50% improvements in sample complexity over PPO for difficult performance thresholds, demonstrating the practical benefits of counterfactual baselines in temporally extended environments.

Collectively, these findings confirm that SCM-based counterfactual baselines constitute a powerful and theoretically principled tool for stabilizing gradient-based RL. The advantages persist across both bandit and MDP settings, and the combination with PPO proves especially effective.

Future Work

While this thesis provides a detailed exploration of SCM-driven variance reduction, several promising directions remain for future research.

- 1. Experiments in More Realistic or High-Dimensional Environments.** Future investigations could evaluate counterfactual baselines in settings with richer temporal structure, such as robotics locomotion, wireless communication simulations, or resource management tasks with large state spaces. Integrating SCM-based baselines with deep neural policies in high-dimensional continuous control remains an open and challenging direction.
- 2. Learning the SCM or Counterfactual Model from Data.** This work assumes access to the underlying SCM, but many real-world scenarios lack an explicit causal model. Extending these ideas to learned causal models, causal representation learning, or hybrid generative models (e.g., diffusion models trained to simulate counterfactuals) could greatly broaden the applicability of counterfactual RL.
- 3. Explainability and Interpretability of Causal Interventions.** SCMs provide a natural foundation for interpretability: nodes correspond to semantically meaningful variables and interventions provide mechanistic explanation. A systematic study of how RL agents behave under counterfactual queries, which actions most influence outcomes, and how noise propagates through causal pathways could yield valuable explainability tools for RL.
- 4. Stability and Optimization of Top- k and Probability-Reward Baselines.** Empirical results showed that certain hybrid baselines (e.g., probability-reward weighted variants) can be unstable due to amplification of stochasticity. Further theoretical work could analyze the bias-variance characteristics of these baselines, determine optimal weighting schemes, or design adaptive top- k selection mechanisms tied to policy entropy or uncertainty estimates.
- 5. Extensions to Multi-Agent and Game-Theoretic Settings.** Many queueing and communication systems involve multiple decision-makers whose actions interact through shared causal structure. Extending counterfactual baselines to multi-agent RL—especially settings where agents must reason over the effect of their actions on others’ outcomes—represents a promising direction with important real-world implications.
- 6. Integration with Model-Based RL and Planning.** Because SCMs explicitly encode transition structure, they suggest natural connections to model-based RL. Counterfactual

baselines could be combined with planning methods, causal model predictive control, or uncertainty-aware causal rollouts to further reduce sample complexity.

Closing Remarks

This thesis demonstrates that counterfactual inference offers a powerful lens through which to improve reinforcement learning. By grounding gradient estimation in causal reasoning and leveraging structural knowledge of noise and interventions, we obtain estimators that are more stable, more sample-efficient, and more interpretable. The empirical results, particularly those involving C-PPO, highlight the practical value of integrating causal structure into modern policy-optimization pipelines.

Causal RL remains an emerging field with substantial unexplored potential. The ideas developed here—variance reduction via SCM baselines, causal advantage estimators, and hybrid PPO frameworks—provide a foundation for future research at the intersection of causality and reinforcement learning. It is our hope that these contributions will serve as a step toward more robust, data-efficient, and explainable learning systems.

References

- [1] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*. MIT Press, 2018.
- [2] R. J. Williams, “Simple statistical gradient-following algorithms for connectionist reinforcement learning,” *Machine learning*, vol. 8, no. 3, pp. 229–256, 1992.
- [3] J. Schulman, S. Levine, P. Moritz, M. Jordan, and P. Abbeel, “Trust region policy optimization,” in *ICML*, 2015.
- [4] J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel, “High-dimensional continuous control using generalized advantage estimation,” in *ICLR*, 2016.
- [5] M. Papini, D. Binaghi, G. Canonaco, M. Pirodda, and M. Restelli, “Stochastic variance-reduced policy gradient,” in *International conference on machine learning*, pp. 4026–4035, PMLR, 2018.
- [6] C.-A. Cheng, X. Yan, and B. Boots, “Trajectory-wise control variates for variance reduction in policy gradient methods,” in *Conference on Robot Learning*, pp. 1379–1394, PMLR, 2020.
- [7] M. Gargiani, A. Zanelli, A. Martinelli, T. Summers, and J. Lygeros, “Page-pg: A simple and loopless variance-reduced policy gradient method with probabilistic gradient estimation,” in *International Conference on Machine Learning*, pp. 7223–7240, PMLR, 2022.
- [8] J. Pearl, *Causality*. Cambridge University Press, 2009.
- [9] T. He, J. Gajcin, and I. Dusparic, “Causal counterfactuals for improving the robustness of reinforcement learning,” *arXiv preprint arXiv:2211.05551*, 2022.
- [10] Z. Deng, J. Jiang, G. Long, and C. Zhang, “Causal reinforcement learning: A survey,” *arXiv preprint arXiv:2307.01452*, 2023.

- [11] C.-C. Lu, Y. Jeong, and E. Bareinboim, “Causal reinforcement learning,” in *NeurIPS*, 2021.
- [12] C. Sommer, K. Zhang, and E. Bareinboim, “Permutation-based causal discovery in offline reinforcement learning,” in *NeurIPS*, 2020.
- [13] B. Liu, Q. Xie, and E. Modiano, “Rl-qn: A reinforcement learning framework for optimal control of queueing systems,” *ACM Transactions on Modeling and Performance Evaluation of Computing Systems*, vol. 7, no. 1, pp. 1–35, 2022.
- [14] D. Talbi and Z. Gal, “Integrating reinforcement learning into m/m/1/k retry queueing models for 6g applications,” *Sensors*, vol. 25, no. 12, p. 3621, 2025.
- [15] Y. Sun, M. Peng, and Y. Zhou, “Deep reinforcement learning for resource allocation in wireless networks,” in *IEEE ICC*, 2019.
- [16] L. Wan *et al.*, “Reinforcement learning-based scheduling in wireless systems,” in *IEEE Globecom*, 2020.